

Dyadic Sensitivity and Polynomial-Chaos Frontiers for the Fabius–Rvachev Law

Exact active-digit ANOVA, marked Legendre coefficients,
 q -Pochhammer interaction bounds, and Mellin phase limits

Repository-aware mathematical investigation
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Abstract

The existing Fabius–Rvachev corpus develops the bounded Fabius distribution, Rvachev’s compactly supported density, exact dyadic arithmetic, Thue–Morse cancellation, infinite sinc products, q -analogues, Bell–Bernoulli moment calculus, Legendre expansions, and Lambert- W endpoint asymptotics. This report develops a different layer: the exact sensitivity and orthogonal-chaos structure of observables of the independent dyadic digits underlying the Rvachev law.

For the centered random series

$$X = \sum_{j \geq 1} 2^{-j} V_j, \quad V_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1],$$

we prove an explicit infinite Hoeffding decomposition of e^{tX} . The normalized variance of each component indexed by a finite set S factorizes as $\prod_{j \in S} r(x_j)$, where $x_j = t2^{-j}$ and r is the continuous extension of $x \coth x - 1$ with $r(0) = 0$. Consequently the entire hierarchy of variance shares is the law of a countable Poisson–binomial active-digit process with probabilities

$$p(x) = \frac{r(x)}{1 + r(x)}, \quad p(x) = 1 - \frac{\tanh x}{x} \quad (x \neq 0), \quad p(0) = 0.$$

The same decomposition refines into a tensor Legendre chaos whose local degree marks are given exactly by modified spherical Bessel functions. For an arbitrary smooth observable $g(X)$, a mixed-replacement identity yields the sharp order- k bound

$$\sum_{|S|=k} \|g_S\|_2^2 \leq \frac{\|g^{(k)}\|_\infty^2}{3^k} \frac{4^{-k(k+1)/2}}{(1/4; 1/4)_k},$$

with equality at the top order for monomials. This supplies a new direct bridge from sensitivity analysis to q -Pochhammer arithmetic.

We also derive a two-variable chaos product, Bell-polynomial formulae for its coefficients, a q -binomial small-parameter expansion, and a Mellin analysis of the effective interaction order. At large t that order is not governed by a central limit theorem: after centering by $\lfloor \log_2 t \rfloor$ it converges along dyadic phases to an explicit integer-valued law, while its mean has a real-analytic logarithmically periodic correction. The no-active-digit atom is identified exactly with the negative Fabius Laplace product, thereby transporting the corpus’s endpoint phase to an ANOVA event. A final section shows how the highest sign interaction is a continuous Thue–Morse corner difference and gives a Lambert- W degree cutoff for the Legendre chaos.

All assertions marked proved are supplied with proofs. Numerical tables and figures are reproducible from the fully commented Python program included in the archive. “New” always means not located in the audited repository corpus and screened prior packages; no unconditional claim of worldwide historical priority is made.

Symbol	proof status used throughout
P	proved in this report
I	imported from the audited repository or standard literature
C	conjectural frontier
N	numerical evidence only

Contents

1	Scope, corpus audit, and contribution boundary	4
1.1	What was read and how duplication was screened	4
1.2	Imported material not relabeled as new	4
1.3	Contribution map	5
2	The dyadic law and normalization dictionary	5
2.1	Fabius and Rvachev variables	5
2.2	Moments, cumulants, and Bell polynomials	6
2.3	Product probability space and projections	6
3	Exact Hoeffding decomposition of the exponential observable	7
3.1	The chaos generating product	8
3.2	Newton sums and a Bell-polynomial form	9
4	The active-digit Poisson–binomial law	9
4.1	Exact Sobol indices and finite-coordinate projections	10
4.2	Elementary local estimates	11
5	Marked tensor-Legendre chaos	12
6	A sharp interaction bound for every smooth observable	14
6.1	A mixed-replacement identity	14
6.2	The exact dyadic and geometric-q factors	15
6.3	Sharpness for monomials	16
6.4	Exact polynomial components and Bell–Bernoulli tails	17
6.5	Superexponential order truncation	18
7	Small-field q-binomial asymptotics and Bell–Bernoulli structure	18
7.1	Local expansion and the leading Gaussian-binomial row	18
7.2	The total energy as a Bernoulli–Bell series	20
8	Effective interaction dimension and a Mellin phase law	20
8.1	Refinement and convergent small-field series	20
8.2	Mellin transform of the activation profile	21
8.3	The logarithmically periodic mean	22
8.4	A phase-indexed limiting distribution	24
9	The no-active atom and the negative Fabius Laplace endpoint	26
10	Thue–Morse corners inside the continuous digit chaos	27
11	Lambert-W cutoffs for Legendre chaos	28

12 Reproducible exact and numerical experiments	30
12.1 Certified truncation policy	30
12.2 What the program verifies	30
12.3 Reproduction command	31
13 Conjectures and frontier research program	31
13.1 Mode locking in the phase law	31
13.2 Differential algebra of the chaos products	31
13.3 The full geometric-q phase diagram	32
13.4 Inverse-Fabius observables	32
13.5 Best N-term and downward-closed chaos approximation	32
13.6 Strict log-concavity and local limit theory	32
13.7 Automatic signs in the sign–magnitude refinement	33
13.8 Formalization roadmap	33
14 Conclusion	33
A Status and dependency matrix	33
B Compact formula atlas	34
C Corpus nonduplication ledger	35
D Archive manifest and reproducibility boundary	36

1 Scope, corpus audit, and contribution boundary

1.1 What was read and how duplication was screened

The requested source is a living mathematical corpus rather than one paper. The audit covered the public documentation tree under `Analysis/FabiusFunction/docs`, its current primary exposition, the current consolidated frontier sources, imported paper transcriptions, frozen historical revisions, and a preserved recursive path ledger. The ledger records 78 visible `*.tex` paths at its 27 August 2026 checkpoint; a later consolidation uses a smaller claim-level count because generated table fragments and absorbed historical variants are not independent mathematical sources. The difference is a counting convention, not a coverage claim. The complete preserved ledger is included in `audit/corpus_manifest_2026-08-27.txt`.

The current controlling mathematical interfaces were read at theorem, proof, result-ledger, and open-problem level. They include:

1. the primary Fabius–Rvachev exposition and its normalization dictionary;
2. the Thue–Morse and signed-summation consolidations;
3. exponent-sequence, q -Pochhammer, Gaussian-binomial, and inverse- q developments;
4. Fourier-decay, zero-divisor, spectral, and Pascal–Rvachev volumes;
5. inverse-Fabius, Lambert- W , Bell, and saddle-expansion volumes;
6. Legendre, Lagrange, interpolation, sampling, transform, and operator representations;
7. imported classical papers and archived provenance sources;
8. previously generated reports in the user’s library, used as a negative control against accidental rediscovery.

Exact and close-variant searches were run for the organizing phrases *Hoeffding decomposition*, *functional ANOVA*, *Sobol index*, *Efron–Stein*, *polynomial chaos*, *active digit*, *Esscher tilt*, and *chi-square divergence*, together with formula searches for $x \coth x - 1$ and $1 - \tanh x/x$ in a Fabius/Rvachev context. No repository treatment of the theorem network developed below was located. A targeted literature search found the general theories of Hoeffding projections, global sensitivity, polynomial chaos, and Mellin harmonic sums, but no exact Fabius–Rvachev specialization matching the results here.

Priority convention

Every novelty statement in this report is repository-relative. It means that the result was not found in the audited current corpus, its indexed historical sources, or the prior generated packages screened as negative controls. The classical ingredients—Hoeffding projections, Legendre polynomials, Poisson–binomial laws, Mellin transforms, q -Pochhammer symbols, and Lambert W —are standard and are cited. Their exact assembly for the dyadic Fabius–Rvachev law is the contribution considered here.

1.2 Imported material not relabeled as new

The following facts are inputs from the repository and classical literature:

- the random-uniform representation of the Fabius distribution and the centered Rvachev density;
- the functional-differential equations and the infinite sinc/hyperbolic products;
- exact dyadic Fabius values, Thue–Morse finite splines, and q -binomial formulae;
- Bernoulli-number cumulants and complete Bell-polynomial moments;
- Legendre expansions of the one-dimensional density;

- the exact negative-Laplace product and its quadratic-plus-periodic logarithmic asymptotic;
- inverse-Fabius Lambert- W expansions, spectral zero arithmetic, Pascal–Rvachev hierarchies, and previously developed operator, information-geometric, free/Boolean, zonoid, and projection theories.

1.3 Contribution map

Repository-relative theorem network

The proved contributions are:

1. an exact infinite Hoeffding decomposition of e^{tX} and all component norms;
2. a Poisson–binomial law for the active set and exact first/total Sobol indices;
3. a marked tensor-Legendre refinement and a local Parseval identity;
4. a sharp mixed-replacement inequality for smooth observables, its exact dyadic q -Pochhammer factor, and the monomial top order;
5. exact polynomial component formulae in terms of tail Bell–Bernoulli moments;
6. a chaos generating product with Newton–Bell recurrences and a q -binomial small- t expansion;
7. a Mellin transform for the local activation profile, an explicit log-periodic mean, and a phase-indexed limiting law for interaction order;
8. an exact identification of the zero-active atom with the negative Fabius Laplace product;
9. a Thue–Morse corner identity for highest sign interactions;
10. a Lambert- W degree cutoff for local Legendre-chaos truncation.

2 The dyadic law and normalization dictionary

2.1 Fabius and Rvachev variables

Let $U_0, U_1, \dots$ be independent $\text{Unif}[0, 1]$ variables and put

$$Y = \sum_{j \geq 0} 2^{-j-1} U_j. \quad (1)$$

The bounded Fabius function F is the distribution function of Y . Set $V_j = 2U_{j-1} - 1$ for $j \geq 1$ and

$$X = 2Y - 1 = \sum_{j \geq 1} a_j V_j, \quad a_j = 2^{-j}, \quad V_j \stackrel{\text{iid}}{\sim} \text{Unif}[-1, 1]. \quad (2)$$

Then X has Rvachev density up on $[-1, 1]$, and

$$\text{up}(x) = F(1 - |x|), \quad F(y) = \text{up}(y - 1), \quad F'(y) = 2 \text{up}(2y - 1) \quad (0 \leq y \leq 1) \quad (3)$$

up to the repository's stated convention for the affine density/CDF bridge. We use the centered law because its independent coordinates and symmetry are transparent.

For $x \in \mathbb{C}$, define

$$m(x) = \frac{\sinh x}{x}, \quad m(0) = 1. \quad (4)$$

The moment generating function of X is

$$M(t) = \mathbb{E}e^{tX} = \prod_{j \geq 1} m(t2^{-j}). \quad (5)$$

The product converges locally uniformly and satisfies

$$M(2t) = m(t)M(t). \quad (6)$$

On the imaginary axis, $M(2\pi iz)$ becomes the repository's normalized infinite sinc product after the corresponding Fourier convention is applied.

2.2 Moments, cumulants, and Bell polynomials

Write κ_n for the cumulants of X . The corpus gives

$$\kappa_{2r} = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}, \quad \kappa_{2r+1} = 0, \quad (7)$$

where B_n are Bernoulli numbers. Hence

$$\mathbb{E}X^n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n), \quad (8)$$

with $\mathcal{B}_n$ the complete exponential Bell polynomial. These formulae are imported; below they become coefficients of sensitivity and chaos objects.

2.3 Product probability space and projections

Let $\Omega = [-1, 1]^{\mathbb{N}}$ with product probability measure $\mu = \bigotimes_{j \geq 1} (dv_j/2)$. For each coordinate, let

$$E_j f = \int_{-1}^1 f(v_1, \dots, v_{j-1}, u, v_{j+1}, \dots) \frac{du}{2}, \quad Q_j = I - E_j. \quad (9)$$

For a finite set $S \subset \mathbb{N}$, let $\mathcal{F}_S = \sigma(V_j : j \in S)$ and put

$$Q_S = \prod_{j \in S} Q_j, \quad E_{S^c} f = \mathbb{E}[f \mid \mathcal{F}_S], \quad f_S = Q_S E_{S^c} f. \quad (10)$$

Thus E_{S^c} means integration over all complementary coordinates, not a literal unordered infinite product of operators. Equivalently it is the L^2 limit of the finite complementary-coordinate integrations. This conditional-expectation definition makes the infinite product notation exact. The commuting orthogonal projections give the Hoeffding or functional-ANOVA expansion

$$f = \sum_{S \in \mathbb{N}} f_S, \quad \|f\|_2^2 = \sum_{S \in \mathbb{N}} \|f_S\|_2^2, \quad (11)$$

for $f \in L^2(\mu)$, with convergence in L^2 . Here $S \in \mathbb{N}$ means that S is finite. The empty component is $f_\emptyset = \mathbb{E}f$.

Remark 2.1. The report uses the term *digit* for the continuous independent coordinates V_j . They are not binary digits. Binary and Thue–Morse signs enter later through a Rademacher decomposition of each uniform coordinate.

3 Exact Hoeffding decomposition of the exponential observable

Fix $t \in \mathbb{R}$ and write

$$x_j = ta_j = t2^{-j}, \quad Z_j(t) = \frac{e^{x_j V_j}}{m(x_j)} - 1. \quad (12)$$

Then $\mathbb{E}Z_j(t) = 0$ and

$$e^{tX} = M(t) \prod_{j \geq 1} (1 + Z_j(t)). \quad (13)$$

The next theorem turns this elementary product into the complete orthogonal sensitivity decomposition.

Theorem 3.1 (Exact dyadic Hoeffding components; **P**). *For every finite $S \subset \mathbb{N}$, the Hoeffding component of e^{tX} indexed by S is*

$$H_S(t) = M(t) \prod_{j \in S} Z_j(t). \quad (14)$$

The series

$$e^{tX} = \sum_{S \in \mathbb{N}} H_S(t) \quad (15)$$

converges unconditionally in $L^2(\mu)$, its terms are mutually orthogonal, and

$$\|H_S(t)\|_2^2 = M(t)^2 \prod_{j \in S} r(x_j), \quad r(x) = \frac{m(2x)}{m(x)^2} - 1. \quad (16)$$

Here $r(0) = 0$, and $r(x) = x \coth x - 1$ for $x \neq 0$; thus r is defined and analytic at the origin as well. The empty product gives $H_\emptyset(t) = M(t)$.

Proof. Let $\mathcal{F}_n = \sigma(V_1, \dots, V_n)$ and

$$M_{>n}(t) = \prod_{j > n} m(x_j).$$

The finite-prefix conditional expectation, including its indispensable tail normalization, is

$$f_n := \mathbb{E}[e^{tX} \mid \mathcal{F}_n] = M_{>n}(t) e^{t \sum_{j=1}^n a_j V_j} = M(t) \prod_{j=1}^n (1 + Z_j).$$

Applying E_j selects the constant term in coordinate j , while Q_j selects Z_j . Thus for $S \subset \{1, \dots, n\}$ the component of f_n indexed by S is already exactly $M(t) \prod_{j \in S} Z_j$, independently of n .

For the infinite limit, first compute

$$\begin{aligned} \mathbb{E}Z_j^2 &= \frac{m(2x_j)}{m(x_j)^2} - 1 \\ &= \frac{m(2x_j)}{m(x_j)^2} - 1 = r(x_j), \end{aligned}$$

where the identity with $x_j \coth x_j - 1$ holds when $x_j \neq 0$, and the displayed quotient gives its value 0 at the origin. The Taylor expansion $r(x) = x^2/3 + O(x^4)$ implies $\sum_j r(x_j) < \infty$. Hence

$$\prod_{j \geq 1} (1 + r(x_j)) < \infty.$$

The products in (14) are orthogonal because every pair of distinct sets contains a coordinate occurring in only one product, and that coordinate has mean zero. Their squared norms therefore sum to

$$M(t)^2 \sum_{S \in \mathbb{N}} \prod_{j \in S} r(x_j) = M(t)^2 \prod_{j \geq 1} (1 + r(x_j)) < \infty.$$

Thus the orthogonal series converges unconditionally in L^2 . Its prefix sums are the conditional expectations f_n , which converge in L^2 to e^{tX} by the L^2 martingale convergence theorem (and also pointwise by the convergent random series for X). This proves the identity and the norm formula. $\square$

Corollary 3.2 (Second-moment product and refinement quotient; **P**). *Define*

$$R(t) = \frac{M(2t)}{M(t)^2}. \quad (17)$$

Then

$$R(t) = \prod_{j \geq 1} (1 + r(t2^{-j})) = \frac{m(t)}{M(t)}. \quad (18)$$

Consequently

$$\text{Var}(e^{tX}) = M(t)^2 (R(t) - 1). \quad (19)$$

Proof. Parseval in theorem 3.1 gives $M(2t) = M(t)^2 \prod_j (1 + r_j)$. The second identity follows from $M(2t) = m(t)M(t)$ in (6). Removing the empty component gives the variance. $\square$

3.1 The chaos generating product

For $k \geq 0$, define the unnormalized order- k coefficient

$$A_k(t) = \sum_{\substack{S \subseteq \mathbb{N} \\ |S|=k}} \prod_{j \in S} r(t2^{-j}), \quad A_0(t) = 1, \quad (20)$$

and the bivariate product

$$\mathcal{C}(t, z) = \prod_{j \geq 1} (1 + zr(t2^{-j})) = \sum_{k \geq 0} A_k(t) z^k. \quad (21)$$

Thus $M(t)^2 A_k(t)$ is the total L^2 energy at interaction order k .

Proposition 3.3 (Dyadic chaos refinement; **P**). *The product satisfies*

$$\mathcal{C}(2t, z) = (1 + zr(t)) \mathcal{C}(t, z), \quad (22)$$

and hence

$$A_k(2t) = A_k(t) + r(t)A_{k-1}(t), \quad k \geq 1. \quad (23)$$

Moreover,

$$A_1(t) = \sum_{j \geq 1} r(t2^{-j}) = t \frac{d}{dt} \log M(t). \quad (24)$$

Proof. Separate the $j = 1$ factor after replacing t by $2t$:

$$\mathcal{C}(2t, z) = \prod_{j \geq 1} (1 + zr(t2^{1-j})) = (1 + zr(t)) \prod_{j \geq 1} (1 + zr(t2^{-j})).$$

Coefficient comparison gives (23). Finally,

$$x \frac{d}{dx} \log m(x) = x \coth x - 1 = r(x) \quad (x \neq 0),$$

with both outer members extending to 0 at the origin. Termwise differentiation of the locally uniformly convergent product for M gives (24). $\square$

3.2 Newton sums and a Bell-polynomial form

Put

$$s_\ell(t) = \sum_{j \geq 1} r(t2^{-j})^\ell, \quad \ell \geq 1. \quad (25)$$

Absolute convergence follows from $r(x) = O(x^2)$ at zero.

Proposition 3.4 (Newton–Bell formula; **P**). *For every $k \geq 1$,*

$$A_k(t) = \frac{1}{k!} \mathcal{B}_k(s_1(t), -1!s_2(t), \dots, (-1)^{k-1}(k-1)!s_k(t)). \quad (26)$$

The power sums obey

$$s_\ell(2t) = s_\ell(t) + r(t)^\ell. \quad (27)$$

Proof. Taking the logarithm of (21) gives

$$\log \mathcal{C}(t, z) = \sum_{\ell \geq 1} \frac{(-1)^{\ell-1}}{\ell} s_\ell(t) z^\ell.$$

The exponential formula for complete Bell polynomials yields (26). The refinement of s_ℓ is obtained by the same index shift as in [proposition 3.3](#). $\square$

4 The active-digit Poisson–binomial law

The local factor r has a canonical conversion from odds to probability:

$$p(x) = \frac{r(x)}{1+r(x)} \quad (x \in \mathbb{R}), \quad p(x) = 1 - \frac{\tanh x}{x} \quad (x \neq 0), \quad p(0) = 0. \quad (28)$$

For real $x \neq 0$, one has $0 < p(x) < 1$. Let

$$B_j(t) \sim \text{Bernoulli}(p(t2^{-j})) \quad (29)$$

be independent, and define the finite almost-sure count

$$K_t = \sum_{j \geq 1} B_j(t). \quad (30)$$

Finiteness follows from $p(x) \leq x^2/3$, proved below.

Theorem 4.1 (Variance shares are an active-set law; **P**). *For every finite set $S \subset \mathbb{N}$,*

$$\frac{\|H_S(t)\|_2^2}{\mathbb{E}e^{2tX}} = \mathbb{P}(\{j : B_j(t) = 1\} = S). \quad (31)$$

For $t \neq 0$, the normalized nonconstant Hoeffding energy is therefore

$$\frac{\|H_S(t)\|_2^2}{\text{Var}(e^{tX})} = \mathbb{P}(\{j : B_j(t) = 1\} = S \mid K_t > 0), \quad S \neq \emptyset. \quad (32)$$

In particular, the order- k variance share is

$$\frac{M(t)^2 A_k(t)}{\text{Var}(e^{tX})} = \mathbb{P}(K_t = k \mid K_t > 0). \quad (33)$$

Proof. Since $p_j = r_j/(1 + r_j)$ and $1 - p_j = 1/(1 + r_j)$,

$$\mathbb{P}(\{j : B_j = 1\} = S) = \prod_{j \in S} \frac{r_j}{1 + r_j} \prod_{j \notin S} \frac{1}{1 + r_j} = \frac{\prod_{j \in S} r_j}{R(t)}.$$

Use $\mathbb{E}e^{2tX} = M(t)^2 R(t)$ and (16). Conditioning away the empty set and summing over $|S| = k$ prove the remaining identities. $\square$

Corollary 4.2 (No-active atom; **P**). *The probability of no active digit is*

$$\mathbb{P}(K_t = 0) = \frac{1}{R(t)} = \frac{M(t)^2}{M(2t)} = \frac{M(t)}{m(t)}. \quad (34)$$

Equivalently,

$$R(t) - 1 = \chi^2(\mathbb{P}_t \| \mathbb{P}), \quad \frac{d\mathbb{P}_t}{d\mathbb{P}} = \frac{e^{tX}}{M(t)}, \quad (35)$$

so the active-set law decomposes the chi-square divergence of the Esscher tilt into orthogonal digit interactions.

Proof. The first identity is the empty-set case of [theorem 4.1](#); the other forms use (18). The chi-square identity is

$$\mathbb{E} \left(\frac{e^{tX}}{M(t)} - 1 \right)^2 = \frac{M(2t)}{M(t)^2} - 1 = R(t) - 1.$$

$\square$

4.1 Exact Sobol indices and finite-coordinate projections

For $f_t = e^{tX}$, define the first-order and total-effect Sobol indices by

$$S_j^{(1)}(t) = \frac{\|H_{\{j\}}(t)\|_2^2}{\text{Var}(f_t)}, \quad S_j^T(t) = \frac{\sum_{S \ni j} \|H_S(t)\|_2^2}{\text{Var}(f_t)}. \quad (36)$$

Corollary 4.3 (Closed Sobol formulae; **P**). *For $t \neq 0$,*

$$S_j^{(1)}(t) = \frac{r(t2^{-j})}{R(t) - 1}, \quad S_j^T(t) = \frac{p(t2^{-j})}{1 - \mathbb{P}(K_t = 0)}. \quad (37)$$

More generally, if $\mathcal{F}_m = \sigma(V_1, \dots, V_m)$ and

$$R_m(t) = \prod_{j=1}^m (1 + r(t2^{-j})), \quad (38)$$

then

$$\|\mathbb{E}[f_t | \mathcal{F}_m]\|_2^2 = M(t)^2 R_m(t), \quad (39)$$

$$\|f_t - \mathbb{E}[f_t | \mathcal{F}_m]\|_2^2 = M(t)^2 (R(t) - R_m(t)). \quad (40)$$

Proof. The first-order formula is immediate. For the total effect,

$$\sum_{S \ni j} \prod_{i \in S} r_i = r_j \prod_{i \neq j} (1 + r_i) = p_j R.$$

Divide by $R - 1$ and use $1 - \mathbb{P}(K_t = 0) = (R - 1)/R$. Conditional expectation retains exactly the components with $S \subset \{1, \dots, m\}$. Orthogonality then gives (39) and (40). $\square$

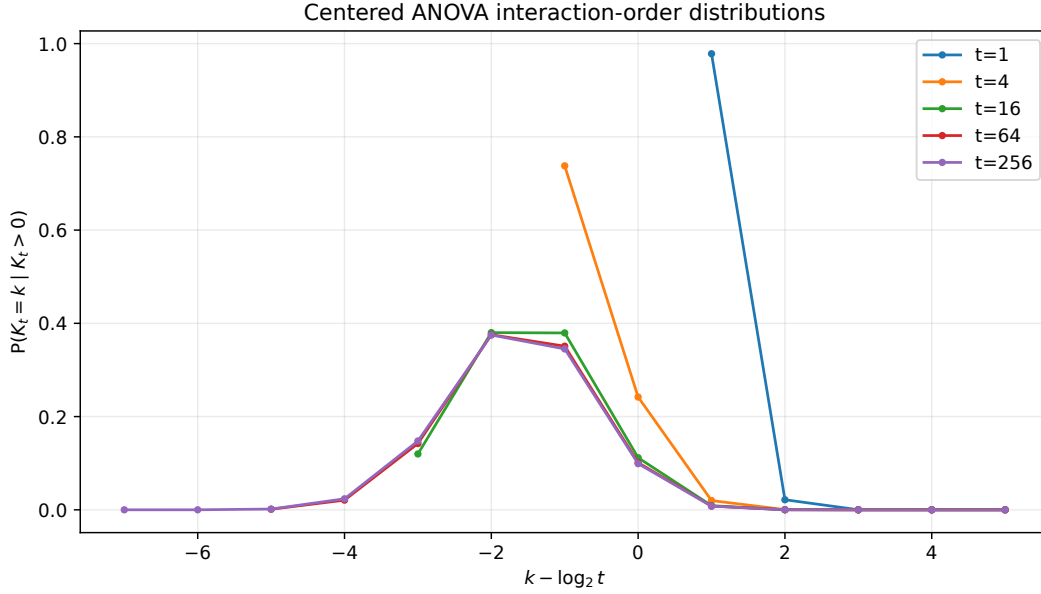


Figure 1: Exact Poisson–binomial order weights $\mathbb{P}(K_t = k \mid K_t > 0)$ for several values of t . As t increases, the interaction order translates roughly by $\log_2 t$ rather than widening on a central-limit scale.

4.2 Elementary local estimates

Lemma 4.4 (Activation bounds; **P**). *For $x \geq 0$,*

$$0 \leq p(x) \leq \min\left(1, \frac{x^2}{3}\right), \quad (41)$$

For $x > 0$ one also has

$$0 < 1 - p(x) = \frac{\tanh x}{x} \leq \min\left(1, \frac{1}{x}\right), \quad 1 - p(0) = 1. \quad (42)$$

Also

$$p(x) = \frac{x^2}{3} - \frac{2x^4}{15} + \frac{17x^6}{315} - \frac{62x^8}{2835} + O(x^{10}). \quad (43)$$

Proof. The lower bounds and $p < 1$ follow from $0 < \tanh x < x$ for $x > 0$, with the origin supplied by the continuous definitions in (28). The inequality $\tanh x \geq x - x^3/3$ follows because the derivative of $\tanh x - x + x^3/3$ is $x^2 - \tanh^2 x \geq 0$. This proves $p(x) \leq x^2/3$. Since $\tanh x \leq 1$, the second upper bound follows. The series is obtained from the standard Taylor series of $\tanh x/x$. $\square$

Corollary 4.5 (Uniform interaction-count variance; **P**). *For all $t > 0$,*

$$\text{Var}(K_t) = \sum_{j \geq 1} p(t2^{-j})(1 - p(t2^{-j})) \leq \frac{22}{9}. \quad (44)$$

Proof. For $0 < t < 1$, use $p(x) \leq x^2/3$ to obtain $\text{Var}(K_t) \leq \sum_j t^2/(3 \cdot 4^j) = t^2/9 < 1/9$. For $t \geq 1$, put $m = \lfloor \log_2 t \rfloor$. On $j \leq m$ use $p(1 - p) \leq 1 - p \leq 2^j/t$; hence the sum is less than 2. On $j > m$, use $p(1 - p) \leq p \leq t^2/(3 \cdot 4^j)$, whose tail is at most $4/9$. The stated uniform bound follows. $\square$

5 Marked tensor-Legendre chaos

The one-dimensional Legendre expansion refines each active coordinate into a local polynomial degree. Let

$$\ell_n(v) = \sqrt{2n+1} P_n(v), \quad n \geq 0, \quad (45)$$

so that $\{\ell_n\}_{n \geq 0}$ is an orthonormal basis of $L^2([-1, 1], dv/2)$. Define the modified spherical-Bessel coefficient

$$i_n(x) = \frac{1}{2} \int_{-1}^1 e^{xv} P_n(v) dv. \quad (46)$$

Thus $i_0(x) = m(x)$ and the classical plane-wave/Legendre expansion reads

$$e^{xv} = \sum_{n \geq 0} (2n+1) i_n(x) P_n(v) = \sum_{n \geq 0} \sqrt{2n+1} i_n(x) \ell_n(v). \quad (47)$$

For a finitely supported multi-index $\nu = (\nu_1, \nu_2, \dots) \in \mathbb{N}_0^{(\mathbb{N})}$, put

$$\ell_\nu(V) = \prod_{j \geq 1} \ell_{\nu_j}(V_j), \quad \text{supp } \nu = \{j : \nu_j > 0\}. \quad (48)$$

Proposition 5.1 (Local active-degree Parseval identity; **P**). *For every real x ,*

$$\boxed{\sum_{n \geq 1} (2n+1) \frac{i_n(x)^2}{i_0(x)^2} = r(x)}. \quad (49)$$

For $x \neq 0$, define the local degree probabilities

$$q_n(x) = \frac{(2n+1) i_n(x)^2}{m(2x) - m(x)^2}, \quad n \geq 1. \quad (50)$$

They sum to one and extend continuously to the origin by

$$q_1(0) = 1, \quad q_n(0) = 0 \quad (n \geq 2). \quad (51)$$

Consequently the local degree generating function has the unambiguous continuous definition

$$D(x, z) = \begin{cases} \sum_{n \geq 1} q_n(x) z^n, & x \neq 0, \\ z, & x = 0, \end{cases} \quad |z| \leq 1. \quad (52)$$

The convergence to $D(0, z) = z$ is uniform for $|z| \leq 1$.

Proof. Parseval applied to (47) gives

$$m(2x) = \frac{1}{2} \int_{-1}^1 e^{2xv} dv = \sum_{n \geq 0} (2n+1) i_n(x)^2.$$

Subtract the $n = 0$ term and divide by $m(x)^2$. This proves (49); for $x \neq 0$, the denominator in (50) is the same positive nonconstant local energy. Finally, $i_n(x) = x^n / (2n+1)!! + O(x^{n+2})$ and $m(2x) - m(x)^2 = x^2/3 + O(x^4)$. Hence $q_1(x) \rightarrow 1$, while $\sum_{n \geq 2} q_n(x) = 1 - q_1(x) \rightarrow 0$. This proves (51) and the asserted uniform convergence on the closed unit disk. $\square$

Theorem 5.2 (Exact tensor-Legendre coefficients; **P**). *The expansion*

$$e^{tX} = \sum_{\nu \in \mathbb{N}_0^{(N)}} c_\nu(t) \ell_\nu(V) \quad (53)$$

converges in L^2 , with

$$c_\nu(t) = M(t) \prod_{j \in \text{supp } \nu} \frac{\sqrt{2\nu_j + 1} i_{\nu_j}(x_j)}{i_0(x_j)}. \quad (54)$$

For every finite S , summing the squared coefficients with support exactly S recovers the Hoeffding component:

$$\sum_{\text{supp } \nu = S} |c_\nu(t)|^2 = \|H_S(t)\|_2^2. \quad (55)$$

For $t \neq 0$, the normalized nonconstant squared tensor coefficients are therefore generated by independent active indicators B_j and, on active coordinates, independent marks N_j with $\mathbb{P}(N_j = n \mid B_j = 1) = q_n(x_j)$, conditioned on at least one activation. At zero local field the continuous convention is $N_j = 1$; it is harmless because then $B_j = 0$ almost surely.

Proof. Local Parseval is already established in [proposition 5.1](#). Now fix N and use the exactly normalized conditional expectation

$$f_N = \mathbb{E}[e^{tX} \mid \mathcal{F}_N] = M_{>N}(t) \prod_{j=1}^N e^{x_j V_j}.$$

Expanding these N local factors by (47) gives a finite tensor expansion in each finite degree truncation; completeness and local Parseval give its L^2 limit. Since $M_{>N}(t) \prod_{j=1}^N i_0(x_j) = M(t)$, its coefficient for every ν with $\text{supp } \nu \subset \{1, \dots, N\}$ is exactly (54).

The martingale f_N converges in L^2 to e^{tX} . The nested finite tensor bases therefore yield (53) and show that the coefficient squares sum to $\|e^{tX}\|_2^2 = M(2t)$. A tensor coefficient lies in the Hoeffding subspace indexed by S exactly when $\text{supp } \nu = S$; Parseval inside that closed subspace proves (55). Dividing the local nonconstant energies by their sums gives the independent mark law, and [theorem 4.1](#) supplies the independent active indicators. $\square$

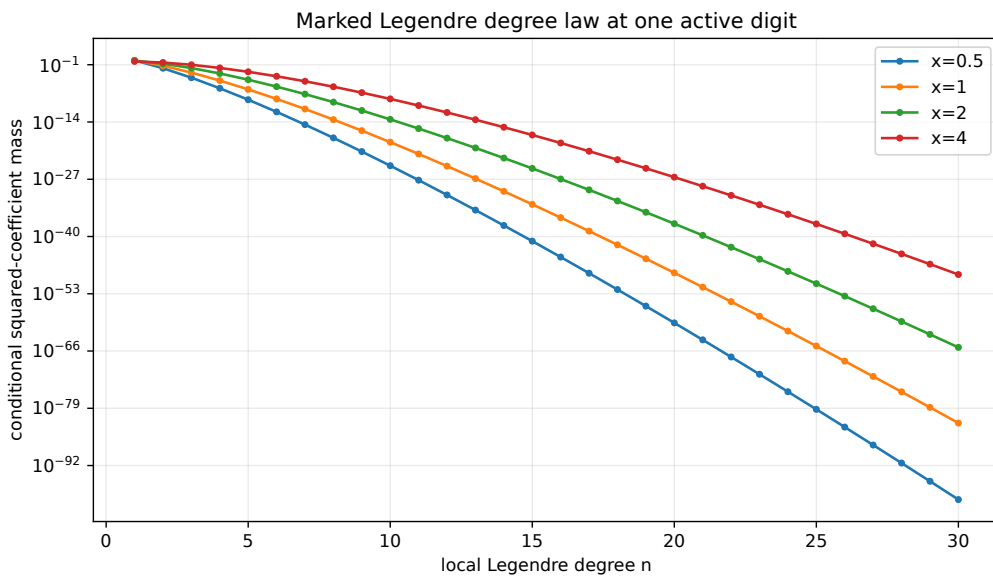


Figure 2: The local degree mark in (50). Larger local field x activates higher Legendre degrees, while the far dyadic tail remains almost entirely degree one when it activates.

Corollary 5.3 (Joint interaction-degree generating function; **P**). *Let u mark active-coordinate count and z mark total Legendre degree. The normalized second-moment generating product, for $|u|, |z| \leq 1$, is*

$$\frac{1}{M(2t)} \sum_{\nu} |c_{\nu}(t)|^2 u^{|\text{supp } \nu|} z^{|\nu|_1} = \prod_{j \geq 1} [(1 - p_j) + p_j u D_j(z)], \quad (56)$$

where $D_j(z) = D(x_j, z)$, so in particular $D_j(z) = z$ when $x_j = 0$. Equivalently, for $x_j \neq 0$,

$$D_j(z) = \sum_{n \geq 1} \frac{(2n+1)i_n(x_j)^2}{m(2x_j) - m(x_j)^2} z^n. \quad (57)$$

Conditioning away the all-zero multi-index gives the corresponding variance law.

Proof. This is the probability-generating function of the independent marked process identified by theorems 4.1 and 5.2 and proposition 5.1. $\square$

6 A sharp interaction bound for every smooth observable

The exponential case is exactly factorized, but the dyadic geometry controls much more general nonlinear observables. Let $g: [-1, 1] \rightarrow \mathbb{R}$ and $f(V) = g(X)$, with X as in (2). Write g_S for the Hoeffding component of f .

6.1 A mixed-replacement identity

For a finite set S , let V'_j be independent copies of V_j , independent of the original sequence. If h depends on V_S , define

$$\Delta_S h(V_S, V'_S) = \prod_{j \in S} (I - T'_j) h(V_S), \quad (58)$$

where T'_j replaces V_j by V'_j .

Lemma 6.1 (Exact replacement norm; **P**). *For every $h \in L^2(\sigma(V_j : j \in S))$,*

$$\mathbb{E} |\Delta_S h(V_S, V'_S)|^2 = 2^{|S|} \left\| \prod_{j \in S} Q_j h \right\|_2^2. \quad (59)$$

Proof. For one coordinate,

$$\mathbb{E} |h(V) - h(V')|^2 = 2\mathbb{E} |h - \mathbb{E}h|^2 = 2\|Qh\|_2^2.$$

Apply this identity successively to the coordinates in S . At each step the replacement difference commutes with all projections in the other coordinates, and contributes a factor 2 while applying the corresponding Q_j . Induction gives (59). $\square$

Theorem 6.2 (Mixed-derivative interaction inequality; **P**). *Let $S \subseteq \mathbb{N}$ with $|S| = k$, and suppose $g \in C^k([-1, 1])$. Then*

$$\|g_S\|_2^2 \leq \frac{\|g^{(k)}\|_{\infty}^2}{3^k} \prod_{j \in S} a_j^2. \quad (60)$$

Consequently, the total order- k energy

$$\mathcal{E}_k(g) = \sum_{\substack{S \subseteq \mathbb{N} \\ |S|=k}} \|g_S\|_2^2 \quad (61)$$

satisfies

$$\mathcal{E}_k(g) \leq \frac{\|g^{(k)}\|_\infty^2}{3^k} e_k(a_1^2, a_2^2, \dots), \quad (62)$$

where e_k is the k th elementary symmetric function.

Proof. Set

$$h_S(v_S) = \mathbb{E}[g(X) \mid V_S = v_S].$$

Then $g_S = \prod_{j \in S} Q_j h_S$. Differentiation under the expectation gives

$$\frac{\partial^k h_S}{\prod_{j \in S} \partial v_j}(v_S) = \left(\prod_{j \in S} a_j \right) \mathbb{E}[g^{(k)}(X) \mid V_S = v_S]. \quad (63)$$

Repeated use of the fundamental theorem of calculus over the rectangle with opposite corners V_S and V'_S yields

$$|\Delta_S h_S| \leq \|g^{(k)}\|_\infty \prod_{j \in S} a_j |V_j - V'_j|.$$

By lemma 6.1,

$$\begin{aligned} \|g_S\|_2^2 &\leq 2^{-k} \|g^{(k)}\|_\infty^2 \prod_{j \in S} a_j^2 \mathbb{E}(V_j - V'_j)^2 \\ &= 2^{-k} \|g^{(k)}\|_\infty^2 \prod_{j \in S} a_j^2 \left(\frac{2}{3}\right)^k, \end{aligned}$$

which is (60). Summing over $|S| = k$ proves (62). $\square$

6.2 The exact dyadic and geometric- q factors

The elementary symmetric function in (62) has a closed q -Pochhammer form.

Lemma 6.3 (Geometric elementary symmetric function; **I**). *For $0 < Q < 1$,*

$$e_k(Q, Q^2, Q^3, \dots) = \frac{Q^{k(k+1)/2}}{(Q; Q)_k}, \quad (64)$$

$$e_k(1, Q, Q^2, \dots) = \frac{Q^{k(k-1)/2}}{(Q; Q)_k}, \quad (65)$$

where $(Q; Q)_k = \prod_{r=1}^k (1 - Q^r)$.

Proof. The q -binomial theorem gives

$$\prod_{j \geq 1} (1 + zQ^j) = \sum_{k \geq 0} \frac{Q^{k(k+1)/2}}{(Q; Q)_k} z^k.$$

Coefficient comparison proves (64); adjoining the $j = 0$ factor gives (65). $\square$

Corollary 6.4 (Dyadic q -Pochhammer interaction bound; **P**). *For the Fabius–Rvachev weights $a_j = 2^{-j}$,*

$$\mathcal{E}_k(g) \leq \frac{\|g^{(k)}\|_\infty^2}{3^k} \frac{4^{-k(k+1)/2}}{(1/4; 1/4)_k}. \quad (66)$$

More generally, for

$$X_q = (1 - q) \sum_{j \geq 0} q^j V_j, \quad 0 < q < 1, \quad (67)$$

whose coefficients sum to one,

$$\mathcal{E}_{q,k}(g) \leq \frac{\|g^{(k)}\|_\infty^2}{3^k} \frac{(1-q)^{2k} q^{k(k-1)}}{(q^2; q^2)_k}. \quad (68)$$

Proof. Apply lemma 6.3 with $Q = 1/4$ in the dyadic case and with $Q = q^2$ after factoring $(1-q)^{2k}$ in the geometric- q case. $\square$

Remark 6.5 (What is sharp). The constant 3^{-k} and the complete elementary-symmetric dependence on the coordinate amplitudes cannot be reduced uniformly over all degree- k polynomials: theorem 6.6 below attains the bound at the highest possible interaction order. For a fixed nonpolynomial observable, the inequality need not be equality because the mixed derivative varies across the replacement rectangle.

6.3 Sharpness for monomials

Theorem 6.6 (Exact top-order monomial energy; **P**). *Let $g(x) = x^d$ with $d \geq 1$. Then $g_S = 0$ whenever $|S| > d$. If $|S| = d$, then*

$$g_S(V_S) = d! \prod_{j \in S} a_j V_j \quad (69)$$

and

$$\|g_S\|_2^2 = \frac{(d!)^2}{3^d} \prod_{j \in S} a_j^2. \quad (70)$$

Consequently

$$\mathcal{E}_d(x^d) = \frac{(d!)^2}{3^d} e_d(a_1^2, a_2^2, \dots), \quad (71)$$

so (62) is exact at $k = d$. For the dyadic law,

$$\mathcal{E}_d(x^d) = \frac{(d!)^2}{3^d} \frac{4^{-d(d+1)/2}}{(1/4; 1/4)_d}. \quad (72)$$

Proof. Expand $X^d = (\sum_j a_j V_j)^d$. A monomial in the coordinates involves at most d distinct indices, so no Hoeffding component of order above d can occur. If $|S| = d$, the only term involving every coordinate in S is the multilinear term $d! \prod_{j \in S} a_j V_j$; every other term either omits an index or has total degree greater than d . Since $\mathbb{E}V_j = 0$, this multilinear term is already pure in every coordinate. Its squared norm is $(d!)^2 \prod_{j \in S} a_j^2 \mathbb{E}V_j^2 = (d!)^2 3^{-d} \prod a_j^2$. Summing and using lemma 6.3 proves the result. $\square$

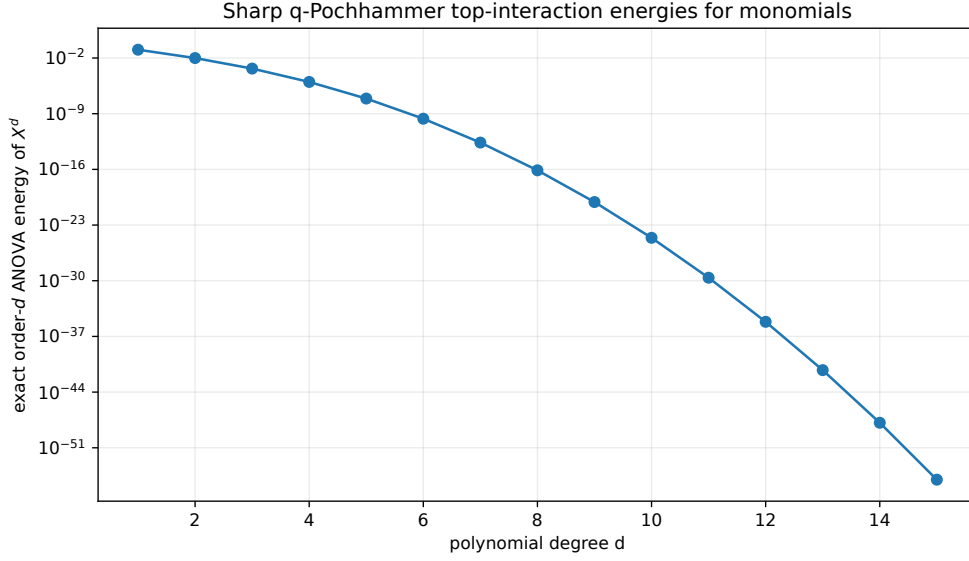


Figure 3: The exact highest-order interaction energy (72). The quadratic exponent in the dyadic factor forces superexponential decay even after multiplication by $(d!)^2$.

6.4 Exact polynomial components and Bell–Bernoulli tails

For a finite S , write

$$X_{S^c} = \sum_{j \notin S} a_j V_j, \quad m_r^{(S^c)} = \mathbb{E} X_{S^c}^r, \quad \mu_r = \mathbb{E} V_1^r = \frac{1 + (-1)^r}{2(r+1)}. \quad (73)$$

Proposition 6.7 (All monomial Hoeffding components; **P**). *For $g(x) = x^d$ and a nonempty finite S ,*

$$g_S(V_S) = \sum_{\substack{\alpha_j \geq 1 \ (j \in S) \\ |\alpha| \leq d}} \frac{d! m_{d-|\alpha|}^{(S^c)}}{(d-|\alpha|)! \prod_{j \in S} \alpha_j!} \prod_{j \in S} a_j^{\alpha_j} (V_j^{\alpha_j} - \mu_{\alpha_j}). \quad (74)$$

The tail moments are complete Bell polynomials in the tail cumulants:

$$m_r^{(S^c)} = \mathcal{B}_r(\kappa_1^{(S^c)}, \dots, \kappa_r^{(S^c)}), \quad \kappa_r^{(S^c)} = \sum_{j \notin S} a_j^r \kappa_r(V_1). \quad (75)$$

For dyadic rational amplitudes, every coefficient in (74) is rational and can be generated exactly from Bernoulli numbers.

Proof. Condition on V_S and expand by the multinomial theorem:

$$\mathbb{E}[X^d | V_S] = \sum_{|\alpha| \leq d} \frac{d! m_{d-|\alpha|}^{(S^c)}}{(d-|\alpha|)! \prod_{j \in S} \alpha_j!} \prod_{j \in S} (a_j V_j)^{\alpha_j}.$$

Applying Q_j annihilates every term with $\alpha_j = 0$ and replaces $V_j^{\alpha_j}$ by $V_j^{\alpha_j} - \mu_{\alpha_j}$ when $\alpha_j \geq 1$. This proves (74). Independence makes cumulants additive and gives (75). The uniform cumulants are rational Bernoulli combinations, and the dyadic amplitudes are rational. $\square$

Example 6.8 (The first two orders). For $g(x) = x^2$,

$$g_{\{j\}} = a_j^2 \left(V_j^2 - \frac{1}{3} \right), \quad g_{\{i,j\}} = 2a_i a_j V_i V_j.$$

For $g(x) = x^3$,

$$g_{\{j\}} = a_j^3 V_j^3 + 3a_j V_j \operatorname{Var}(X_{\{j\}^c}),$$

while for distinct i, j ,

$$g_{\{i,j\}} = 3a_i^2 a_j \left(V_i^2 - \frac{1}{3} \right) V_j + 3a_i a_j^2 V_i \left(V_j^2 - \frac{1}{3} \right),$$

and $g_{\{i,j,k\}} = 6a_i a_j a_k V_i V_j V_k$.

6.5 Superexponential order truncation

Corollary 6.9 (Analytic-observable order tail; **P**). *Suppose that for some $C, A > 0$,*

$$\|g^{(k)}\|_\infty \leq C A^k k! \quad (k \geq 0). \quad (76)$$

Then the dyadic order energies obey

$$\mathcal{E}_k(g) \leq C^2 \left(\frac{A^2}{3} \right)^k (k!)^2 \frac{4^{-k(k+1)/2}}{(1/4; 1/4)_k}, \quad (77)$$

and the truncation after order K satisfies

$$\left\| g(X) - \sum_{|S| \leq K} g_S \right\|_2^2 \leq \sum_{k > K} C^2 \left(\frac{A^2}{3} \right)^k (k!)^2 \frac{4^{-k(k+1)/2}}{(1/4; 1/4)_k}. \quad (78)$$

The right side decays faster than $\exp(-cK^2)$ for every fixed A .

Proof. Insert (76) into (66) and sum the orthogonal tail. Since $(1/4; 1/4)_k$ decreases to the positive infinite product $(1/4; 1/4)_\infty$ and $\log(k!) = O(k \log k)$, the term $4^{-k(k+1)/2}$ dominates by a negative quadratic exponent. $\square$

7 Small-field q -binomial asymptotics and Bell–Bernoulli structure

7.1 Local expansion and the leading Gaussian-binomial row

The local odds, totalized by $r(0) = 0$, have the convergent expansion

$$r(x) = \frac{x^2}{3} - \frac{x^4}{45} + \frac{2x^6}{945} - \frac{x^8}{4725} + O(x^{10}), \quad |x| < \pi. \quad (79)$$

For $x \neq 0$ this is the Taylor expansion of $x \coth x - 1$. Let $Q = 1/4$ and abbreviate

$$E_k(Q) = \frac{Q^{k(k+1)/2}}{(Q; Q)_k}. \quad (80)$$

Theorem 7.1 (Dyadic q -binomial chaos asymptotic; **P**). *For every fixed $k \geq 1$, as $t \rightarrow 0$,*

$$A_k(t) = \frac{t^{2k}}{3^k} \frac{4^{-k(k+1)/2}}{(1/4; 1/4)_k} + O_k(t^{2k+2}). \quad (81)$$

The first correction is

$$A_k(t) = a^k t^{2k} E_k(Q) \quad (82)$$

$$+ b a^{k-1} t^{2k+2} (E_1(Q) E_k(Q) - (k+1) E_{k+1}(Q)) + O_k(t^{2k+4}), \quad (83)$$

where $a = 1/3$, $b = -1/45$, and $Q = 1/4$.

Proof. Write $q_j = Q^j$. From (79),

$$r(t2^{-j}) = at^2q_j + bt^4q_j^2 + O(t^6q_j^3),$$

uniformly in j for t in a fixed compact subinterval of $(-\pi, \pi)$. The leading term of the elementary symmetric function A_k is $a^kt^{2k}e_k(q_1, q_2, \dots) = a^kt^{2k}E_k(Q)$ by lemma 6.3.

At relative order t^2 , exactly one selected factor contributes its b term. The required symmetric sum is

$$T_k = \sum_j q_j^2 e_{k-1}(q_1, \dots, \widehat{q_j}, \dots).$$

Multiplying $e_1 e_k$ and separating whether the index from e_1 already belongs to the k -set gives

$$e_1 e_k = T_k + (k+1)e_{k+1}.$$

This proves (83); absolute geometric summability controls the remainder. $\square$

Corollary 7.2 (General geometric- q leading law; **P**). *For X_q in (67), let $A_{q,k}(t)$ be the order- k coefficient for e^{tX_q} . Then*

$$A_{q,k}(t) \sim \left(\frac{(1-q)^2 t^2}{3} \right)^k \frac{q^{k(k-1)}}{(q^2; q^2)_k} \quad (t \rightarrow 0). \quad (84)$$

Proof. Use $r(x) \sim x^2/3$ and (65) with $Q = q^2$. $\square$

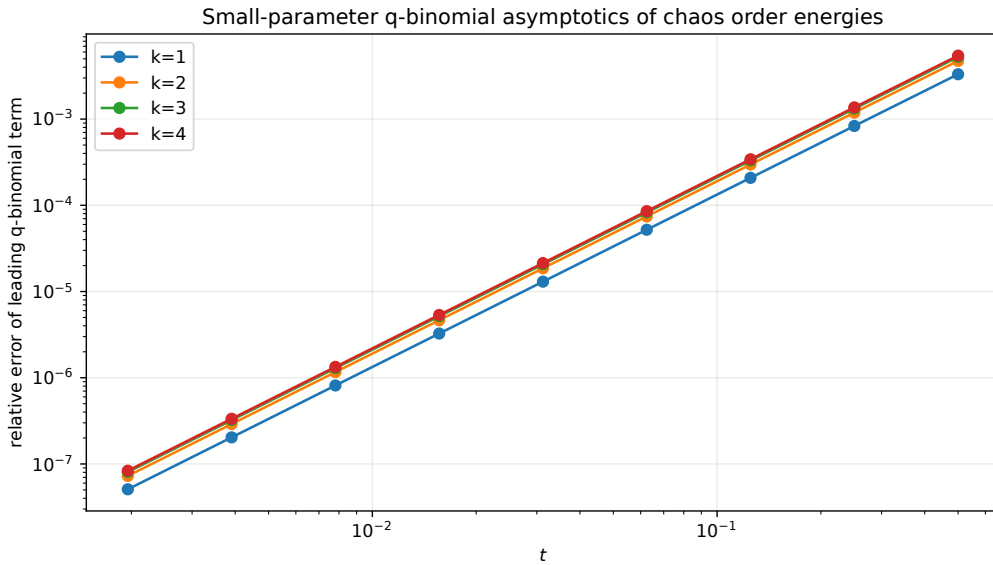


Figure 4: Relative errors of the leading and first-corrected formulae for $A_k(t)$. The first correction in (83) improves the power of t by two, as predicted by the exact symmetric-sum calculation.

Corollary 7.3 (Small-field order law; **P**). *As $t \rightarrow 0$,*

$$\mathbb{P}(K_t = k \mid K_t > 0) \sim 3^{2-k} E_k(1/4) t^{2k-2}, \quad k \geq 1. \quad (85)$$

In particular, the conditional variance energy concentrates at first order, with the order- k deficit carrying the exact Gaussian-binomial coefficient $E_k(1/4)$.

Proof. Since $R - 1 = \sum_{k \geq 1} A_k$ and $A_1(t) \sim t^2 E_1(1/4)/3 = t^2/9$, divide (81) by this leading term. $\square$

7.2 The total energy as a Bernoulli–Bell series

The repository cumulants (7) give an independent exact expansion of the total normalized variance.

Proposition 7.4 (Bell–Bernoulli energy coefficients; **P**). *Define*

$$\lambda_{2m} = (2^{2m} - 2)\kappa_{2m} = \frac{(2^{2m} - 2)2^{2m-1}B_{2m}}{m(4^m - 1)}, \quad m \geq 1, \quad \lambda_{2m+1} = 0, \quad m \geq 0. \quad (86)$$

Then, as Taylor series in the complex variable t ,

$$\log R(t) = \sum_{m \geq 1} \lambda_{2m} \frac{t^{2m}}{(2m)!}, \quad |t| < \pi, \quad (87)$$

$$R(t) = \sum_{n \geq 0} \mathcal{B}_n(\lambda_1, \dots, \lambda_n) \frac{t^n}{n!}, \quad |t| < 4\pi, \quad (88)$$

$$R(t) - 1 = \sum_{m \geq 1} \mathcal{B}_{2m}(0, \lambda_2, 0, \lambda_4, \dots) \frac{t^{2m}}{(2m)!}, \quad |t| < 4\pi. \quad (89)$$

Thus the sum over every nonempty interaction order is encoded by a complete Bell transform of the Bernoulli–Mersenne cumulants.

Proof. The cumulant expansion of $\log M$ gives

$$\log R(t) = \log M(2t) - 2 \log M(t) = \sum_{n \geq 1} (2^n - 2)\kappa_n \frac{t^n}{n!}.$$

Odd terms vanish. Exponentiating by the complete Bell-polynomial formula proves the result near the origin. The displayed domains are the exact Taylor disks. Indeed $\log R$ first meets a zero of $R(t) = m(t)/M(t)$ at $t = \pm\pi i$, whereas the apparent singularities of R at $t = \pm 2\pi i$ are removable: the simple zeros of $m(t)$ and $M(t)$ cancel. At $t = \pm 4\pi i$, $M(t)$ has two zero factors while $m(t)$ has only one, so R has a pole. Thus the logarithmic series has radius π , while the Taylor series for R and $R - 1$ have radius 4π . $\square$

Remark 7.5 (Two Bell structures). There are two distinct Bell transforms in the report. Formula (26) reconstructs each *interaction order* from power sums of the local odds r_j . Formula (88) reconstructs the *sum over all orders* from cumulants of the one-dimensional Rvachev law. Their equality after summing A_k is a nontrivial bridge between local digit sensitivity and global Bernoulli-number moment arithmetic.

8 Effective interaction dimension and a Mellin phase law

The mean number of active coordinates is

$$\mu(t) = \mathbb{E}K_t = \sum_{j \geq 1} p(t2^{-j}). \quad (90)$$

It is distinct from $A_1(t) = \sum r_j$: the former counts the Bernoulli active set underlying normalized coefficient energy, while the latter is the raw first-order elementary symmetric sum.

8.1 Refinement and convergent small-field series

Proposition 8.1 (Effective-dimension refinement; **P**). *For $t \geq 0$,*

$$\boxed{\mu(2t) - \mu(t) = p(t)}. \quad (91)$$

If $|t| < \pi$, then

$$\mu(t) = \sum_{m \geq 1} \frac{c_m}{4^m - 1} t^{2m}, \quad c_m = -\frac{2^{2m+2}(2^{2m+2} - 1)B_{2m+2}}{(2m+2)!}. \quad (92)$$

For the geometric- q law (67),

$$\mu_q(t) = \sum_{j \geq 0} p((1-q)tq^j), \quad (93)$$

$$\mu_q(t) - \mu_q(qt) = p((1-q)t), \quad (94)$$

$$\mu_q(t) = \sum_{m \geq 1} \frac{c_m(1-q)^{2m}}{1-q^{2m}} t^{2m}, \quad |(1-q)t| < \frac{\pi}{2}. \quad (95)$$

Proof. The dyadic refinement is an index shift. The Taylor series $\tanh x/x = 1 - \sum_{m \geq 1} c_m x^{2m}$, with the displayed Bernoulli coefficients, converges for $|x| < \pi/2$. In the dyadic sum the largest local argument is $t/2$, so $|t| < \pi$ suffices. Summing $\sum_{j \geq 1} 4^{-mj} = 1/(4^m - 1)$ gives (92). The same argument with coordinates $(1-q)q^j$ gives the geometric- q identities. $\square$

8.2 Mellin transform of the activation profile

Define, initially in their convergence strips,

$$\mathcal{M}_p(s) = \int_0^\infty p(x)x^{s-1} dx, \quad \mathcal{I}_p(s) = \int_0^\infty p'(x)x^s dx. \quad (96)$$

Theorem 8.2 (Closed Mellin transforms; **P**). For $-2 < \Re s < 0$,

$$\mathcal{M}_p(s) = 2^{2-s}(1 - 2^{2-s})\Gamma(s-1)\zeta(s-1). \quad (97)$$

Write $\mathcal{M}_p^{\text{cont}}$ for the meromorphic continuation supplied by the right side of (97). For $-2 < \Re s < 1$,

$$\mathcal{I}_p(s) = -s\mathcal{M}_p^{\text{cont}}(s) = \frac{2^{2-s}(1 - 2^{2-s})\Gamma(s+1)\zeta(s-1)}{1-s}. \quad (98)$$

The right sides are understood after filling removable singularities. On $-2 < \Re s < 0$, $\mathcal{M}_p^{\text{cont}}$ agrees with the defining Mellin integral $\mathcal{M}_p$. On the boundary line $\Re s = 0$ and away from $s = 0$, it equals $-\mathcal{I}_p(s)/s$; the defining integral for $\mathcal{M}_p$ is not absolutely convergent there and does not have an ordinary improper limit. By contrast, the defining integral for $\mathcal{I}_p$ remains absolutely convergent on that line.

Proof. By lemma 4.4, $p(x) = O(x^2)$ at zero and $p(x) = 1 + O(x^{-1})$ at infinity, giving the strip $-2 < \Re s < 0$ for $\mathcal{M}_p$. Also $p'(x) = O(x)$ at zero and $p'(x) = x^{-2} + O(e^{-2x}/x)$ at infinity, giving $-2 < \Re s < 1$ for $\mathcal{I}_p$.

Put $J(s) = \int_0^\infty x^{s-1} \text{sech}^2 x dx$, which is holomorphic on the half-plane $\Re s > 0$. The exponential expansion

$$\text{sech}^2 x = 4 \sum_{n \geq 1} (-1)^{n-1} n e^{-2nx}$$

may be integrated term by term absolutely when $\Re s > 2$, and there gives

$$J(s) = 2^{2-s}\Gamma(s)(1 - 2^{2-s})\zeta(s-1).$$

Both sides are holomorphic on $\Re s > 0$ after filling their removable singularities. The identity theorem therefore extends this equality from $\Re s > 2$ to the whole half-plane $\Re s > 0$; no conditionally justified Fubini interchange is needed.

Now let $0 < \Re s < 1$. Since

$$p'(x) = \frac{\tanh x - x \operatorname{sech}^2 x}{x^2},$$

integration by parts in the $\tanh x$ term is legitimate at both endpoints and gives

$$\mathcal{I}_p(s) = \frac{s}{1-s} J(s).$$

This proves (98) on $0 < \Re s < 1$, and analytic continuation extends it to the full convergence strip of the left side. Finally, integration by parts in (96) gives $\mathcal{I}_p(s) = -s\mathcal{M}_p(s)$ for $-2 < \Re s < 0$: the boundary term $p(x)x^s$ vanishes because $p(x) = O(x^2)$ at zero and $p(x) = 1 + O(x^{-1})$ at infinity. The gamma recurrence $\Gamma(s) = (s-1)\Gamma(s-1)$ gives (97). $\square$

8.3 The logarithmically periodic mean

For $\theta \in \mathbb{R}$, define the absolutely convergent renormalized sum

$$P(\theta) = \sum_{k \leq 0} [p(2^{\theta-k}) - 1] + \sum_{k \geq 1} p(2^{\theta-k}). \quad (99)$$

The estimates in lemma 4.4 imply exponential convergence at both ends.

Theorem 8.3 (Exact dyadic phase expansion; **P**). *Let $t = 2^{n+\theta}$ with $n \in \mathbb{Z}$, $0 \leq \theta < 1$, and $t \geq 1$. Then*

$$\boxed{\mu(t) = n + P(\theta) + \varepsilon(t), \quad 0 \leq \varepsilon(t) \leq \frac{2}{t}.} \quad (100)$$

Moreover $P(\theta + 1) = P(\theta) + 1$. Hence

$$Q(\theta) = P(\theta) - \theta \quad (101)$$

is one-periodic and real analytic, and

$$\boxed{\mu(t) = \log_2 t + Q(\{\log_2 t\}) + O(t^{-1}).} \quad (102)$$

Proof. Reindex $j = n + k$ in (90):

$$\mu(t) - n = \sum_{k=1-n}^0 [p(2^{\theta-k}) - 1] + \sum_{k \geq 1} p(2^{\theta-k}).$$

Completing the first sum to $k = -\infty$ gives (100) with

$$\varepsilon(t) = \sum_{k \leq -n} [1 - p(2^{\theta-k})].$$

Since $1 - p(x) = \tanh x/x \leq 1/x$,

$$\varepsilon(t) \leq \sum_{k \leq -n} 2^{k-\theta} = \frac{2}{t}.$$

An index shift proves $P(\theta + 1) = P(\theta) + 1$. Termwise differentiation is permitted by exponential convergence of the differentiated tails; the Fourier formula below gives real analyticity. $\square$

Let $L = \log 2$, A be Glaisher's constant in the normalization

$$\log A = \frac{1}{12} - \zeta'(-1), \quad (103)$$

and set $\tau_n = 2\pi n/L$.

Theorem 8.4 (Fourier coefficients of the phase; **P**). *With the convention*

$$Q(\theta) = \sum_{n \in \mathbb{Z}} \widehat{Q}(n) e^{2\pi i n \theta}, \quad (104)$$

one has

$$\widehat{Q}(0) = \frac{11}{6} + \frac{\gamma - 12 \log A}{\log 2}, \quad (105)$$

$$\widehat{Q}(n) = \frac{\mathcal{M}_p^{\text{cont}}(-i\tau_n)}{\log 2} = \frac{\mathcal{I}_p(-i\tau_n)}{2\pi i n}, \quad n \neq 0. \quad (106)$$

In (106), the equal expression using $\mathcal{I}_p$ is an absolutely convergent integral. The coefficients decay exponentially; in particular the Fourier series and all its termwise derivatives converge absolutely in a complex strip. Numerically,

$$\widehat{Q}(0) = -1.6404426943766166621038729 \dots, \quad (107)$$

$$\widehat{Q}(1) = (1.7087561142062823 + 1.3093920503208334i)10^{-6}, \quad (108)$$

$$2|\widehat{Q}(1)| = 4.3055104223684092 \times 10^{-6}. \quad (109)$$

Proof. Differentiate (99) and unfold the logarithmic lattice. For $n \neq 0$,

$$\begin{aligned} \widehat{Q}'(n) &= \int_0^1 L \sum_{k \in \mathbb{Z}} 2^{\theta-k} p'(2^{\theta-k}) e^{-2\pi i n \theta} d\theta \\ &= \int_{-\infty}^{\infty} e^y p'(e^y) e^{-i\tau_n y} dy = \int_0^{\infty} p'(x) x^{-i\tau_n} dx = \mathcal{I}_p(-i\tau_n). \end{aligned}$$

Since $\widehat{Q}'(n) = 2\pi i n \widehat{Q}(n)$, the nonzero formula follows from [theorem 8.2](#).

For the mean, the Mellin transform of the harmonic sum is

$$\int_0^{\infty} \mu(t) t^{s-1} dt = \frac{2^s}{1-2^s} \mathcal{M}_p(s), \quad -2 < \Re s < 0. \quad (110)$$

At $s = 0$,

$$\mathcal{M}_p^{\text{cont}}(s) = -\frac{1}{s} + c_0 + O(s), \quad c_0 = \gamma - 12 \log A + \frac{7}{3} \log 2.$$

Indeed, use $\Gamma(s-1) = -s^{-1} + \gamma - 1 + O(s)$, $\zeta(s-1) = -1/12 + s\zeta'(-1) + O(s^2)$, and $2^{2-s}(1-2^{2-s}) = -12 + 28s \log 2 + O(s^2)$. Also

$$\frac{2^s}{1-2^s} = -\frac{1}{s \log 2} - \frac{1}{2} + O(s).$$

The double pole produces $\log_2 t$, and its constant term is $c_0/\log 2 - 1/2$, which simplifies to (105). Finally, Stirling's estimate for $\Gamma(-1 - i\tau_n)$ in (97) gives exponential decay in $|n|$. The numerical values were evaluated at 80 decimal digits by the accompanying program. $\square$

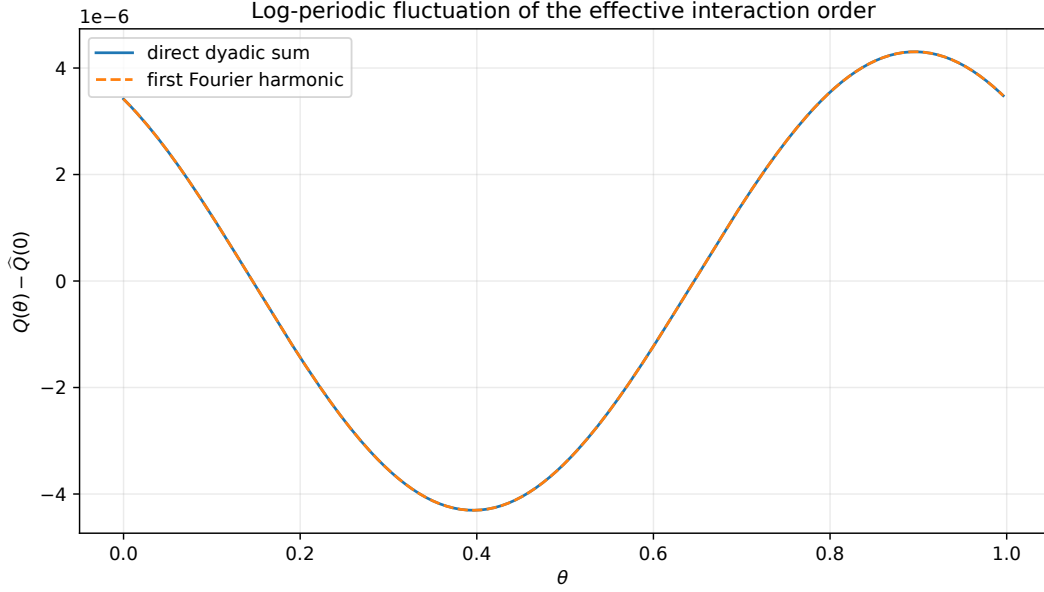


Figure 5: The periodic correction $Q(\theta)$. The direct logarithmic-lattice sum and the first Fourier harmonic are visually indistinguishable at this scale; the total peak amplitude is only about 4.31×10^{-6} .

8.4 A phase-indexed limiting distribution

The bounded variance in [corollary 4.5](#) suggests that the active count does not spread on a Gaussian scale. The exact limit is a two-sided, integer-valued phase law.

For $\theta \in [0, 1)$, let

$$p_{\theta,k} = p(2^{\theta-k}), \quad k \in \mathbb{Z}, \quad (111)$$

and let $B_{\theta,k}$ be independent Bernoulli variables with these parameters. Define

$$K_\theta = \sum_{k \leq 0} (B_{\theta,k} - 1) + \sum_{k \geq 1} B_{\theta,k}. \quad (112)$$

Both sums have only finitely many nonzero terms almost surely: the deficit probabilities $1 - p_{\theta,k}$ are summable on $k \leq 0$, and the activation probabilities $p_{\theta,k}$ are summable on $k \geq 1$.

Theorem 8.5 (Dyadic phase law in total variation; **P**). *Let $t = 2^{n+\theta}$, $0 \leq \theta < 1$, and $n \geq 1$. Then*

$$d_{\text{TV}}(K_t - n, K_\theta) \leq \frac{2}{t}. \quad (113)$$

If J_t denotes the variance-chaos order, that is,

$$J_t \stackrel{\text{d}}{=} K_t \mid (K_t > 0), \quad (114)$$

then

$$d_{\text{TV}}(J_t - n, K_\theta) \leq \frac{4}{t}. \quad (115)$$

For every $z \in \mathbb{C}^\times$, the limiting bilateral probability generating Laurent product is

$$\mathbb{E} z^{K_\theta} = \prod_{k \leq 0} [p_{\theta,k} + (1 - p_{\theta,k})z^{-1}] \prod_{k \geq 1} [(1 - p_{\theta,k}) + p_{\theta,k}z]. \quad (116)$$

Both products converge locally uniformly on $\mathbb{C}^\times$. It has mean

$$\mathbb{E} K_\theta = P(\theta). \quad (117)$$

Proof. Use the same variables $B_{\theta,k}$ to couple the laws. Reindexing the original count gives

$$K_t = \sum_{k \geq 1-n} B_{\theta,k}.$$

Therefore

$$K_t - n = \sum_{k=1-n}^0 (B_{\theta,k} - 1) + \sum_{k \geq 1} B_{\theta,k}.$$

The coupled variables differ only if some omitted coarse index $k \leq -n$ has a deficit $B_{\theta,k} = 0$. By a union bound and $1 - p(x) \leq 1/x$,

$$\mathbb{P}(K_t - n \neq K_\theta) \leq \sum_{k \leq -n} (1 - p_{\theta,k}) \leq \sum_{k \leq -n} 2^{k-\theta} = \frac{2}{t}.$$

This proves (113).

Conditioning a law on an event of probability $1 - \delta$ changes it by total variation exactly δ . Here

$$\mathbb{P}(K_t = 0) = \prod_{j \geq 1} (1 - p(t2^{-j})) \leq 1 - p(t/2) = \frac{\tanh(t/2)}{t/2} \leq \frac{2}{t}.$$

The triangle inequality yields (115). The independent product (116) is immediate from (112). On compact subsets of $\mathbb{C}^\times$, its coarse-factor deviations are bounded by a constant times $1 - p_{\theta,k}$ and its fine-factor deviations by a constant times $p_{\theta,k}$; their sums converge. This proves local uniform convergence. Summing the expectations gives precisely (99). $\square$

Corollary 8.6 (No central-limit widening; **P**). *As $t \rightarrow \infty$, uniformly in the dyadic phase $\{\log_2 t\} \in [0, 1)$,*

$$K_t = \log_2 t + O_{\mathbb{P}}(1), \quad J_t = \log_2 t + O_{\mathbb{P}}(1). \quad (118)$$

Along each fixed phase θ , every uniformly integrable family of powers of $K_{2^{n+\theta}} - n$ has its moment converge to the corresponding moment of K_θ . In particular,

$$\text{Var}(K_{2^{n+\theta}}) \longrightarrow \text{Var}(K_\theta). \quad (119)$$

More precisely, for $t = 2^{n+\theta}$ with $n \geq 1$,

$$0 \leq \text{Var}(K_\theta) - \text{Var}(K_t) = \sum_{k \leq -n} p_{\theta,k}(1 - p_{\theta,k}) \leq \frac{2}{t}. \quad (120)$$

The family obeys the quasi-periodicity

$$K_{\theta+1} \stackrel{d}{=} K_\theta + 1. \quad (121)$$

Proof. Tightness follows from theorem 8.5, or directly from the uniformly bounded variance and the bounded periodic mean correction, with $t \geq 2$ (and hence the stated $t \rightarrow \infty$ interpretation) understood. Under the coupling in theorem 8.5, subtracting n does not change variance and the only terms missing from $K_t - n$ are the independent coarse deficits with $k \leq -n$. Consequently

$$\text{Var}(K_\theta) - \text{Var}(K_t) = \sum_{k \leq -n} p_{\theta,k}(1 - p_{\theta,k}) \leq \sum_{k \leq -n} (1 - p_{\theta,k}) \leq \frac{2}{t},$$

which proves (120) and (119) directly, without inferring uniform integrability of squares from bounded second moments. Reindex $B_{\theta+1,k}$ as $B_{\theta,k-1}$ in (112) to obtain (121). $\square$

Corollary 8.7 (Log-concavity and full support; **P**). *For every phase θ , the mass function of K_θ is log-concave on $\mathbb{Z}$ and strictly positive at every integer. Its set of modes is therefore a nonempty finite interval of consecutive integers.*

Proof. Truncate (112) to $-N \leq k \leq N$. Up to an integer shift, the truncation is a finite Poisson–binomial sum, whose generating polynomial has only real nonpositive zeros; Newton’s inequalities imply log-concavity of its coefficients. The truncations converge in total variation, so the coefficientwise inequalities pass to the limit.

For any prescribed integer, choose finitely many coarse deficits or fine activations producing it. That finite pattern has positive probability, and the probability that no other tail event occurs is a positive convergent product because the relevant tail probabilities are summable. Hence the support is all of $\mathbb{Z}$. A log-concave sequence has a consecutive set of maximizers; because this probability mass tends to zero in both tails, that set is nonempty and finite. Log-concavity alone does not bound its length by two. $\square$

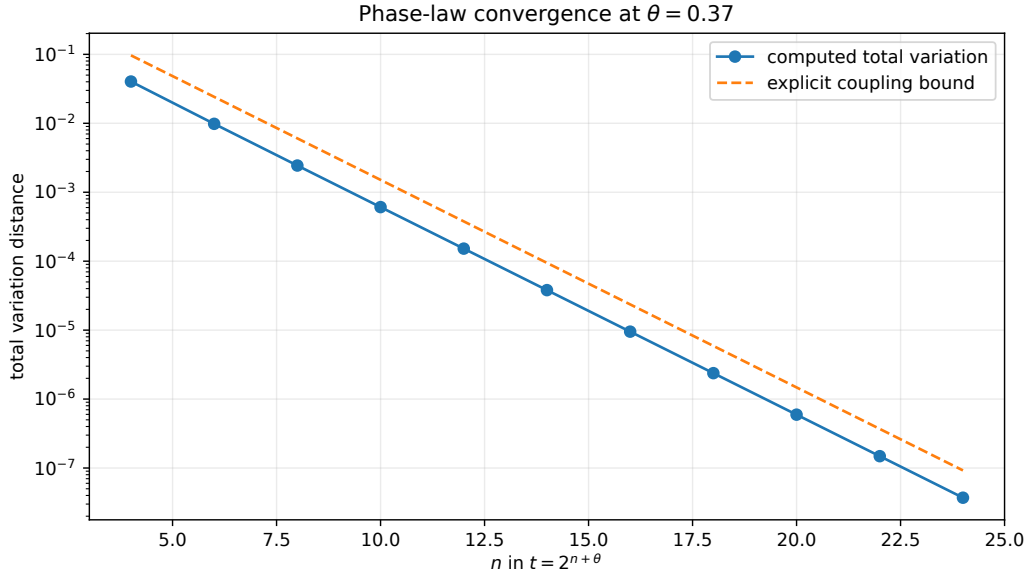


Figure 6: Total-variation convergence at phase $\theta = 0.37$. The computed error decays proportionally to 2^{-n} and remains below the explicit coupling bound $2/t$.

9 The no-active atom and the negative Fabius Laplace endpoint

Let

$$L_F(s) = \mathbb{E}e^{-sY} = \prod_{j \geq 1} \frac{1 - e^{-s/2^j}}{s/2^j}, \quad s \geq 0, \quad (122)$$

be the negative Laplace transform of the Fabius random variable (1). The corpus studies $\Lambda(s) = \log L_F(s)$ in detail, including an exact decomposition into a quadratic logarithmic term, a log-periodic phase, and an exponentially small remainder.

Theorem 9.1 (Exact Laplace transfer of the ANOVA atom; **P**). *For $t > 0$,*

$$\mathbb{P}(K_t = 0) = \frac{2t}{1 - e^{-2t}} L_F(2t). \quad (123)$$

Equivalently,

$$\log \mathbb{P}(K_t = 0) = \log(2t) - \log(1 - e^{-2t}) + \Lambda(2t). \quad (124)$$

Thus every exact or asymptotic statement for the repository’s negative Fabius Laplace product transfers immediately to the probability that the exponential observable has no nonconstant active digit.

Proof. The Fabius law is symmetric: $Y \stackrel{d}{=} 1 - Y$. Therefore the centered variable $X = 2Y - 1$ also satisfies $X \stackrel{d}{=} 1 - 2Y$, and

$$M(t) = \mathbb{E}e^{tX} = e^t \mathbb{E}e^{-2tY} = e^t L_F(2t).$$

Now use (34) and $m(t) = \sinh t/t = (e^t - e^{-t})/(2t)$. $\square$

Corollary 9.2 (Transported quadratic-log phase; **P+I**). *Suppose the repository decomposition is written in the normalization*

$$\Lambda(s) = -\frac{(\log s)^2}{2 \log 2} + \frac{1}{2} \log s + \Psi(\log_2 s) + E(s), \quad (125)$$

where Ψ is one-periodic and $E(s)$ is the explicit exponentially small remainder supplied there. Then

$$\log \mathbb{P}(K_t = 0) = -\frac{(\log(2t))^2}{2 \log 2} + \frac{3}{2} \log(2t) + \Psi(1 + \log_2 t) \quad (126)$$

$$- \log(1 - e^{-2t}) + E(2t). \quad (127)$$

In particular, the same log-periodic endpoint phase controls a concrete high-dimensional sensitivity event.

Proof. Substitute $s = 2t$ from (125) into (124). Periodicity permits replacing $\Psi(1 + \log_2 t)$ by $\Psi(\log_2 t)$ if desired. $\square$

Remark 9.3 (Distinct periodic functions). The periodic correction Q in theorem 8.4 concerns the *mean number* of active digits and has Fourier coefficients governed by the Mellin transform of p . The imported phase Ψ in (125) concerns the *probability of zero* active digits and comes from the Fabius Laplace product. They arise from the same dyadic lattice but are not the same function.

10 Thue–Morse corners inside the continuous digit chaos

The original coordinates are uniform rather than binary. Nevertheless each uniform variable admits the independent sign–magnitude factorization

$$V_j = R_j U_j, \quad R_j \in \{-1, 1\} \text{ Rademacher}, \quad U_j \sim \text{Unif}[0, 1]. \quad (128)$$

This refines the continuous chaos into Walsh sign interactions and magnitude interactions.

For $a \in \mathbb{R}$, let

$$\Delta_a g(x) = g(x + a) - g(x - a). \quad (129)$$

Theorem 10.1 (Continuous Thue–Morse corner identity; **P**). *Let $a_1, \dots, a_N > 0$, let $I \subset \mathbb{R}$ be an open interval containing*

$$\left[x - \sum_{j=1}^N a_j, x + \sum_{j=1}^N a_j \right],$$

and suppose $g \in C^N(I)$. Then

$$\Delta_{a_1} \cdots \Delta_{a_N} g(x) = \sum_{\varepsilon \in \{-1, 1\}^N} \left(\prod_{j=1}^N \varepsilon_j \right) g \left(x + \sum_{j=1}^N a_j \varepsilon_j \right) \quad (130)$$

$$= \int_{-a_1}^{a_1} \cdots \int_{-a_N}^{a_N} g^{(N)}(x + u_1 + \cdots + u_N) du_1 \cdots du_N. \quad (131)$$

For the dyadic amplitudes $a_j = 2^{-j}$,

$$\Delta_{2^{-1}} \cdots \Delta_{2^{-N}} g(x) = (-1)^N \sum_{k=0}^{2^N-1} (-1)^{s_2(k)} g \left(x - (1 - 2^{-N}) + \frac{k}{2^{N-1}} \right), \quad (132)$$

where $s_2(k)$ is the binary digit sum. Hence the corner annihilates every polynomial of degree below N , while

$$\Delta_{2^{-1}} \cdots \Delta_{2^{-N}} x^N = N! 2^{-N(N-1)/2}. \quad (133)$$

Proof. Expanding the product of symmetric differences gives (130). The one-dimensional identity $g(x+a) - g(x-a) = \int_{-a}^a g'(x+u) du$, iterated N times, gives (131).

For dyadic amplitudes, write $\varepsilon_j = 2b_j - 1$ and $k = \sum_{j=1}^N b_j 2^{N-j}$. Then

$$\sum_{j=1}^N 2^{-j} \varepsilon_j = -(1 - 2^{-N}) + \frac{k}{2^{N-1}}, \quad \prod_{j=1}^N \varepsilon_j = (-1)^N (-1)^{s_2(k)}.$$

This proves (132). If $\deg g < N$, then $g^{(N)} = 0$ in (131). For $g(x) = x^N$, the integrand is $N!$ and the box volume is $\prod_j 2a_j = 2^N 2^{-N(N+1)/2} = 2^{-N(N-1)/2}$. $\square$

Corollary 10.2 (Highest Walsh sign coefficient; **P**). *Condition on the magnitudes $U_1, \dots, U_N$ and on a tail variable Z independent of the signs. Then*

$$\mathbb{E}_R \left[g \left(Z + \sum_{j=1}^N a_j R_j U_j \right) \prod_{j=1}^N R_j \right] = 2^{-N} \Delta_{a_1 U_1} \cdots \Delta_{a_N U_N} g(Z). \quad (134)$$

At $U_1 = \cdots = U_N = 1$ and $a_j = 2^{-j}$, the highest sign interaction is the canonical Thue–Morse/Prouhet corner (132).

Proof. The conditional expectation over 2^N equally likely sign vectors is 2^{-N} times (130) with amplitudes $a_j U_j$. $\square$

Remark 10.3. The corner identity explains why Thue–Morse cancellation and high-order functional ANOVA meet naturally but are not identical. Hoeffding projection centers the full uniform coordinate; Walsh projection centers only its sign conditional on its magnitude. The sign–magnitude refinement provides the precise bridge.

11 Lambert-W degree cutoffs for Legendre chaos

The corpus uses Lambert W primarily in endpoint inversion. Polynomial chaos creates a different occurrence: inversion of factorial Legendre-coefficient decay.

Proposition 11.1 (Rodrigues bound for local coefficients; **P**). *For $n \geq 0$ and $x \in \mathbb{C}$,*

$$i_n(x) = \frac{x^n}{2^{n+1} n!} \int_{-1}^1 e^{xv} (1-v^2)^n dv, \quad (135)$$

and hence

$$|i_n(x)| \leq e^{|x|} \frac{|x|^n}{(2n+1)!!}. \quad (136)$$

Proof. Insert Rodrigues' formula $P_n(v) = (2^n n!)^{-1} \frac{d^n}{dv^n} (v^2 - 1)^n$ into (46). Integrating by parts n times produces (135); all boundary terms vanish because $(1-v^2)^n$ has the required endpoint zeros. Next,

$$\int_{-1}^1 (1-v^2)^n dv = \frac{2^{2n+1} (n!)^2}{(2n+1)!}.$$

Using $(2n+1)!! = (2n+1)!/(2^n n!)$ gives (136). $\square$

Corollary 11.2 (Lambert-W inversion rule; **P**). *Fix $x \neq 0$ and define the exact envelope and its monotone-tail threshold by*

$$b_n(x) = e^{|x|} \frac{|x|^n}{(2n+1)!!}, \quad N_x = \left\lceil \frac{|x|}{2} \right\rceil. \quad (137)$$

For every $\varepsilon > 0$, define

$$n_\varepsilon(x) = \min \{n \in \mathbb{N}_0 : n \geq N_x \text{ and } b_n(x) \leq \varepsilon\}. \quad (138)$$

This set is nonempty, and the resulting cutoff is certified for the entire tail:

$$|i_m(x)| \leq b_m(x) \leq \varepsilon \quad \text{for every } m \geq n_\varepsilon(x). \quad (139)$$

As $\varepsilon \downarrow 0$, define

$$A_\varepsilon = \log \frac{e^{|x|}}{\varepsilon}. \quad (140)$$

The degree at which the envelope (136) reaches tolerance ε has the leading inversion

$$n_\varepsilon(x) \sim \frac{A_\varepsilon}{W_0\left(\frac{2A_\varepsilon}{e^{|x|}}\right)}. \quad (141)$$

Thus (141) is an asymptotic initial guess for the exact, tail-valid cutoff (138), not merely for a single coefficient.

Proof. The exact ratio

$$\frac{b_{n+1}(x)}{b_n(x)} = \frac{|x|}{2n+3} < 1 \quad (n \geq N_x)$$

shows that the envelope is decreasing there and tends to zero, so the defining set is nonempty and its minimum certifies every later envelope value. This proves (139) from [proposition 11.1](#).

For the asymptotic, put

$$F_x(n) = \log((2n+1)!!) - n \log |x|,$$

so that $b_n(x) \leq \varepsilon$ is equivalent to $F_x(n) \geq A_\varepsilon$. As $\varepsilon \downarrow 0$ one has $n_\varepsilon(x) \rightarrow \infty$. Hence, for all sufficiently small ε , minimality on the monotone tail gives, with $n = n_\varepsilon(x)$,

$$F_x(n-1) < A_\varepsilon \leq F_x(n).$$

Since

$$F_x(n) - F_x(n-1) = \log \frac{2n+1}{|x|} = O(\log n),$$

this sandwiches the threshold sharply enough to give $F_x(n) = A_\varepsilon + O(\log n)$. Stirling's formula gives

$$\log((2n+1)!!) = n \log \frac{2n}{e} + O(\log n).$$

Therefore the exact minimal cutoff satisfies

$$n \log \frac{2n}{e^{|x|}} = A_\varepsilon + O(\log n).$$

Solving the leading equation with $W_0(z)e^{W_0(z)} = z$ yields (141). Moreover, the derivative of $y \mapsto y \log(2y/(e^{|x|}))$ is $1 + \log(2y/(e^{|x|})) \rightarrow \infty$. Hence the $O(\log n)$ perturbation moves the inverse by $o(n)$ and does not affect the asymptotic ratio. $\square$

Remark 11.3 (Adaptive tensor truncation). At digit j , the local field is $x_j = t2^{-j}$. Formula (141) therefore assigns a different degree cutoff to every active digit. Coarse digits may require several Legendre modes, while far-tail digits have $|x_j| \ll 1$ and their conditional mark law is overwhelmingly concentrated at degree one. This is substantially more efficient than a rectangular tensor-degree cutoff.

12 Reproducible exact and numerical experiments

The archive contains `experiments.py`, a self-contained deterministic program using high-precision arithmetic and standard special-function routines. It does not use Monte Carlo simulation. Its role is to check transcriptions, illustrate limiting laws, and quantify approximation errors; none of the analytic theorems depends on a floating-point experiment.

12.1 Certified truncation policy

Every countable probability sum uses the global inequality $p(x) \leq x^2/3$. If the sum is truncated after digit J , then

$$\sum_{j>J} p(t2^{-j}) \leq \frac{t^2}{3} \sum_{j>J} 4^{-j} = \frac{t^2}{9 \cdot 4^J}. \quad (142)$$

The same bound controls the probability that any omitted Bernoulli activation occurs. Hyperbolic-product tails use the analogous estimate $0 \leq \log m(x) \leq x^2/6$ for real x . The chosen precision is 80 decimal digits; CSV files record every truncation depth and bound.

12.2 What the program verifies

The driver performs the following independent computations:

1. direct logarithmic-lattice evaluation of $Q(\theta)$ and comparison with its Mellin Fourier series;
2. exact Poisson–binomial convolutions for the conditioned chaos-order law;
3. total-variation convergence to a deep finite approximation of the phase law and comparison with the bound $2/t$;
4. elementary-symmetric computation of $A_k(t)$ and comparison with (81) and (83);
5. local Legendre Parseval checks using modified spherical Bessel functions;
6. exact rational evaluation of the monomial top energy (72);
7. independent comparison of the product formula for $\mathbb{P}(K_t = 0)$ with the negative Fabius Laplace product (123).

Table 1: Selected reproducibility checks from `data/numerical_summary.txt`.

Quantity	Computed value
$\widehat{Q}(0)$	$-1.6404426943766166621038729159$
$2 \widehat{Q}(1) $	$4.30551042236840917 \times 10^{-6}$
Maximum direct/Fourier error for Q on the plotted grid	$5.8207661 \times 10^{-11}$
Maximum independent error in (123)	$9.2975205 \times 10^{-26}$
Maximum degree-64 local Parseval error	$6.2172489 \times 10^{-15}$
Maximum relative error of the first-corrected A_k grid	3.2116589×10^{-5}
Phase TV error at $(\theta, n) = (0.37, 24)$	3.7100290×10^{-8}
Theoretical coupling bound at the same point	9.2332142×10^{-8}
Exact top energy $\mathcal{E}_{10}(x^{10})$	$2.4950901155034477 \times 10^{-25}$

Table 2: The no-active atom from two independent products.

t	$\mathbb{P}(K_t = 0)$	$\log \mathbb{P}(K_t = 0)$	absolute discrepancy
1	$8.99201007738952 \times 10^{-1}$	-0.1062486791	9.30×10^{-26}
8	$8.24685731480165 \times 10^{-2}$	-2.4953379898	8.53×10^{-27}
32	$6.44285582709265 \times 10^{-4}$	-7.3473684787	6.66×10^{-29}
128	$3.14592569682258 \times 10^{-7}$	-14.9719874649	3.25×10^{-32}
256	$2.45775445064264 \times 10^{-9}$	-19.8240177288	2.54×10^{-34}

12.3 Reproduction command

From the archive root, run

```
python experiments.py -root .
```

This regenerates all CSV files, the numerical summary, and both vector and raster versions of the figures. Dependencies and tested minimum versions are listed in `requirements.txt`.

Warning 12.1. The Fourier comparison error in [table 1](#) is dominated by the deliberately finite direct-lattice evaluation, not by the four-harmonic Fourier series: the second Fourier harmonic is already of order 10^{-12} and subsequent harmonics are dramatically smaller. Likewise, finite Poisson–binomial calculations are evidence for, not substitutes for, the coupling proof in [theorem 8.5](#).

13 Conjectures and frontier research program

The proved structure opens several questions that do not reduce to the repository’s existing endpoint, spectral, or q -inverse programs.

13.1 Mode locking in the phase law

Conjecture 13.1 (Unique phase mode). *For every $\theta \in [0, 1)$, the full-support log-concave law K_θ has a unique mode except at finitely many phases. At each exceptional phase exactly two adjacent integers tie, and every crossing is transversal.*

Log-concavity from [corollary 8.7](#) reduces this to control of adjacent probability ratios. The quasi-periodicity $K_{\theta+1} \stackrel{d}{=} K_\theta + 1$ suggests one mode transition per period, but the extremely small oscillation of the mean does not by itself prove that picture. A promising route is to differentiate the Laurent product (116) and establish strict monotonicity of the relevant coefficient ratio.

13.2 Differential algebra of the chaos products

Conjecture 13.2 (Differential transcendence). *For generic algebraic $z \neq 0, 1$, neither $t \mapsto \mathcal{C}(t, z)$ nor $t \mapsto \sum_{j \geq 1} p(t2^{-j})$ satisfies a nontrivial algebraic differential equation over $\mathbb{C}(t)$.*

The refinement equations

$$\mathcal{C}(2t, z) = (1 + zr(t))\mathcal{C}(t, z), \quad \mu(2t) - \mu(t) = p(t)$$

place these functions in a multiplicative/additive Mahler-type setting, but the coefficients are hyperbolic rather than rational. One possible strategy is to combine differential-algebraic independence results for shifts with the nonrational zero/pole geometry of $\coth t$ and $\tanh t/t$. A weaker first target is non-D-finiteness.

13.3 The full geometric- q phase diagram

Problem 13.3 (Mellin phase for X_q). *For $0 < q < 1$, determine the exact phase law for the active count associated with X_q in (67), including:*

1. *a total-variation coupling at scale $\log_{1/q} t$;*
2. *the Fourier coefficients of its periodic mean correction;*
3. *uniform asymptotics as $q \rightarrow 1^-$ and $q \rightarrow 0^+$;*
4. *the transition from the almost-continuous $q \rightarrow 1^-$ regime to the sparse lacunary $q \rightarrow 0^+$ regime.*

The dyadic proof extends formally after scaling $t = (1 - q)^{-1}q^{-n-\theta}$, but the dependence of constants and analytic strips on q becomes the central issue. Mellin harmonic-sum theory predicts Fourier frequencies $2\pi n/|\log q|$ and coefficients obtained from $\mathcal{M}_p$ at those frequencies. Making this uniform would connect the present sensitivity theory directly to the corpus's $q \rightarrow 1$ and reciprocal- q programs.

13.4 Inverse-Fabius observables

Problem 13.4 (Quantile-chaos transfer). *Let $G = F^{-1}$ and study the Hoeffding decomposition of nonlinear observables such as*

$$g(X) = G\left(\frac{X+1}{2}\right), \quad g_\alpha(X) = G\left(\frac{X+1}{2}\right)^\alpha.$$

Determine whether the endpoint Lambert- W singularity of G produces a sharp interaction-order law after suitable truncation away from $X = \pm 1$.

The smooth bound (66) becomes delicate because inverse derivatives blow up near the endpoints. A promising decomposition is to separate an interior analytic region from two endpoint layers whose probability is controlled by the known Fabius asymptotics. The resulting balance is expected to contain Lambert W twice: once in the endpoint layer width and once in the Legendre degree cutoff.

13.5 Best N -term and downward-closed chaos approximation

Problem 13.5 (Optimal coefficient geometry). *For fixed t , characterize the sets of multi-indices that capture a prescribed fraction of the variance of e^{tX} with the smallest cardinality. Compare:*

1. *the exact probabilistic ranking generated by the independent marked law;*
2. *downward-closed tensor index sets used in polynomial-chaos algorithms;*
3. *pure interaction-order truncation;*
4. *digit-prefix truncation from (40).*

Because the coefficient-energy measure is explicit, this is a rare infinite-dimensional best- N problem with a fully known product law. It can be reframed as a stochastic knapsack problem with digit activation cost and a conditional local-degree mark.

13.6 Strict log-concavity and local limit theory

Conjecture 13.6 (Strict phase log-concavity). *For every θ , the phase mass function is strictly log-concave except possibly at the unique adjacent pair involved in a mode crossing.*

Finite Poisson–binomial laws are often strictly log-concave under mild nondegeneracy, but taking a two-sided shifted limit can conceal equality cases. A direct proof from the entire Laurent product (116) would also provide quantitative tails and a local limit approximation for the translated order law.

13.7 Automatic signs in the sign–magnitude refinement

Problem 13.7 (Thue–Morse sign topology). *Classify observables g for which the top Walsh sign coefficients in (134), evaluated on dyadic magnitude grids, form automatic or regular sequences. Determine how Prouhet cancellation order, derivative sign changes, and the tensor-Legendre degree marks constrain that sequence.*

This problem uses the corpus’s Thue–Morse machinery but asks a different question: not the value of a finite signed sum itself, but the digital topology of an entire family of continuous-chaos coefficients.

13.8 Formalization roadmap

A staged Lean development can isolate elementary finite statements from analytic limits:

1. **Finite probability algebra.** Formalize coordinate expectation projections, finite Hoeffding orthogonality, the product formula, and the active Bernoulli conversion.
2. **Exact finite combinatorics.** Formalize elementary symmetric functions, the q -binomial coefficient identity, Newton sums, and the monomial sharpness theorem.
3. **Infinite products.** Prove summability from $p(x) \leq x^2/3$ and pass finite decompositions to L^2 limits.
4. **Legendre layer.** Import the orthogonality and local exponential expansion, then prove the marked Parseval law.
5. **Mellin boundary.** Keep the special-function Mellin evaluation as a separate analytic module; the elementary phase coupling and $2/t$ bound do not depend on it.
6. **Repository integration.** Connect the existing Fabius Laplace theorem to (123), preserving the existing trusted endpoint asymptotic as an imported theorem rather than duplicating it.

14 Conclusion

The independent dyadic uniform series has usually been used in the corpus to obtain the Fabius distribution, the Rvachev density, moment formulae, and Fourier products. Treating the same series as an infinite-dimensional input space reveals another exact structure.

For the exponential observable, every Hoeffding component factorizes. The entire variance hierarchy becomes a conditioned Poisson–binomial active-set law, and every active coordinate carries an independent Legendre-degree mark. For arbitrary smooth observables, mixed replacement converts a derivative bound into an exact geometric elementary symmetric function; at dyadic weights this is a q -Pochhammer quotient, and monomials attain the top-order bound. Thus q -series enter not only through finite Fabius values or sinc products, but through the geometry of nonlinear sensitivity itself.

At high field, the interaction order has bounded width around $\log_2 t$. Its phase limit, Mellin transform, and almost imperceptible periodic mean correction are explicit. The zero-active event is exactly the negative Fabius Laplace product, so the already known endpoint phase acquires a new probabilistic meaning. Finally, the sign–magnitude refinement exposes the Thue–Morse corner beneath high-order interactions, while factorial Legendre decay creates a distinct Lambert- W complexity scale.

These results suggest a broader organizing principle: the Fabius–Rvachev law is not only a special smooth distribution but also an exactly solvable model of infinite-dimensional multiscale sensitivity. Its dyadic refinement, q -combinatorics, automatic signs, special-function coefficients, and endpoint phases are different projections of that single product structure.

A Status and dependency matrix

Statement	Status	Main dependency
Exact exponential Hoeffding components	P	independent products and coordinate projections
Totalized component norm $r(0) = 0$	P	one-dimensional hyperbolic integral
Active-set Poisson–binomial law	P	odds-to-probability factorization
First and total Sobol indices	P	active-set law and orthogonality
Tensor-Legendre coefficients	P	standard Legendre exponential expansion
Marked local degree law	P	one-dimensional Parseval identity
Smooth order- k bound	P	mixed replacement and fundamental theorem of calculus
Dyadic q -Pochhammer factor	P	q -binomial theorem
Sharp monomial top order	P	multinomial expansion and uniform variance
All polynomial components	P	conditional multinomial expansion and Bell moments
Small- t q -binomial law	P	local Taylor series and symmetric functions
Bell–Bernoulli total energy	P	repository cumulants and Bell exponential formula
Mellin transform of p	P	Mellin transform of sech^2
Log-periodic mean Q	P	Mellin harmonic-sum analysis
Phase law in total variation	P	explicit Bernoulli coupling
Phase log-concavity	P	finite Poisson–binomial limits
No-active Laplace identity	P	Fabius symmetry and product normalization
Quadratic-log no-active asymptotic	P+I	exact identity plus imported repository endpoint theorem
Thue–Morse corner	P	repeated symmetric differences
Lambert- W degree cutoff	P	Rodrigues bound and Stirling inversion
Unique phase mode	C	strict adjacent-ratio control remains open
Differential transcendence	C	Mahler/differential-algebra methods needed
Inverse-Fabius chaos law	open	endpoint-layer derivative control needed

B Compact formula atlas

Object	Exact formula
Centered law	$X = \sum_{j \geq 1} 2^{-j} V_j, V_j \sim \text{Unif}[-1, 1]$
MGF	$M(t) = \prod_{j \geq 1} m(t2^{-j}), m(0) = 1, m(x) = \sinh(x)/x \text{ for } x \neq 0; M(2t) = m(t)M(t)$
Local Hoeffding odds	$r(x) = m(2x)/m(x)^2 - 1; r(0) = 0 \text{ and } r(x) = x \coth x - 1 \text{ for } x \neq 0$
Local activation probability	$p(x) = r(x)/(1 + r(x)); p(0) = 0 \text{ and } p(x) = 1 - \tanh(x)/x \text{ for } x \neq 0$
Component	$H_S = M \prod_{j \in S} (e^{x_j V_j} / m(x_j) - 1)$
Component energy	$\ H_S\ _2^2 = M^2 \prod_{j \in S} r(x_j)$
Second-moment ratio	$R = M(2t)/M(t)^2 = \prod_j (1 + r_j) = m(t)/M(t)$
No-active atom	$\mathbb{P}(K_t = 0) = 1/R = M(t)/m(t)$
Order product	$\mathcal{C}(t, z) = \prod_j (1 + z r_j) = \sum_k A_k(t) z^k$
Order refinement	$A_k(2t) = A_k(t) + r(t) A_{k-1}(t)$
Power-sum Bell form	$A_k = k!^{-1} \mathcal{B}_k(s_1, -1!s_2, \dots, (-1)^{k-1}(k-1)!s_k)$

Object	Exact formula
Legendre coefficient	$c_\nu = M \prod_{\nu_j > 0} \sqrt{2\nu_j + 1} i_{\nu_j}(x_j)/i_0(x_j)$
Continuous degree mark	$D(x, z) = \sum_{n \geq 1} q_n(x) z^n$ for $x \neq 0$, while $D(0, z) = z$
Smooth order bound	$\mathcal{E}_k(g) \leq 3^{-k} \ g^{(k)}\ _\infty^2 4^{-k(k+1)/2} / (1/4; 1/4)_k$
Monomial top energy	$\mathcal{E}_d(x^d) = (d!)^2 3^{-d} 4^{-d(d+1)/2} / (1/4; 1/4)_d$
Effective mean	$\mu(t) = \sum_j p(t2^{-j})$, $\mu(2t) - \mu(t) = p(t)$
Mellin transform	$\mathcal{M}_p(s) = 2^{2-s} (1 - 2^{2-s}) \Gamma(s-1) \zeta(s-1)$ for $-2 < \Re s < 0$
Periodic mean	$\mu(t) = \log_2 t + Q(\{\log_2 t\}) + O(t^{-1})$
Nonzero Fourier modes	$\widehat{Q}(n) = \mathcal{M}_p^{\text{cont}}(-2\pi i n / \log 2) / \log 2$
Phase coupling	$d_{\text{TV}}(K_t - \lfloor \log_2 t \rfloor, K_\theta) \leq 2/t$
Laplace transfer	$\mathbb{P}(K_t = 0) = 2t(1 - e^{-2t})^{-1} L_F(2t)$
Thue–Morse corner	$\Delta_{2^{-1}} \cdots \Delta_{2^{-N}} g = (-1)^N \sum_{k < 2^N} (-1)^{s_2(k)} g(\cdots)$
Legendre envelope	$ i_n(x) \leq e^{ x } x ^n / (2n+1)!!$
Lambert cutoff	$n_\varepsilon(x) \sim A_\varepsilon / W_0(2A_\varepsilon / (e x))$

C Corpus nonduplication ledger

The audit grouped the recursively inspected sources as follows.

Source family	Material screened for overlap
Primary exposition and glossary	Normalizations, random series, functional equations, moments, cumulants, Fourier products, exact dyadic values, inverse notation, and Legendre data
Lean walkthrough and theorem index	Formalized Appell, q -binomial, endpoint, sampling, derivative, and exact-arithmetic declarations
Consolidated frontier volumes	Bell recurrences, inverse asymptotics, Thue–Morse systems, q -series, spectra, arithmetic, representations, and result registers
Fourier-decay corpus	Dyadic rays, doubling-map martingales, LIL/CLT phenomena, sinc zeros, and sharp product envelopes
Exponent and q -series corpus	General exponent sequences, geometric interpolation, Gaussian binomials, reciprocal q , cyclotomic limits, and inverse q -functions
Inverse and endpoint corpus	Lambert- W branches, all-order Bell coefficients, phase tomography, elasticity, quantile transport, and dyadic inverse values
Integration and operator corpus	Mellin/Laplace/Fourier/resolvent formulae, fractional calculus, Appell deconvolution, self-sampling, Stein–Koopman, and integral identities
Representation corpus	Legendre/Jacobi/Chebyshev expansions, interpolation, cubature, multiresolution, zonoids, and projection theories
Prior generated packages	Pascal spectral probability, free/Boolean cumulants, information geometry, convolution divisors, geometric-comb theory, and related negative controls
Archived papers and historical sources	Classical Fabius/Rvachev papers, Arias de Reyna, Haugland, lacunary-product papers, and frozen repository variants

Targeted searches found no repository treatment of functional ANOVA or Hoeffding projections for the dyadic random series; no Sobol formula for Fabius/Rvachev observables; no active-digit Poisson–binomial variance law; no tensor-Legendre degree marking by local Bessel energy; no smooth interaction bound with the exact $4^{-k(k+1)/2} / (1/4; 1/4)_k$ factor; and no Mellin phase law for effective interaction

order. The audit files in the archive document the search vocabulary, controlling sources, exclusions, and counting caveat.

D Archive manifest and reproducibility boundary

The maintained report package contains (the original ZIP was consumed at repository intake):

Path	Purpose
<code>fabius_dyadic_chaos_frontiers.tex</code>	Complete LaTeX source
<code>fabius_dyadic_chaos_frontiers.pdf</code>	Rendered report
<code>experiments.py</code>	Fully commented deterministic experiment driver
<code>requirements.txt</code>	Python dependencies
<code>README.md</code>	Build and reproduction instructions
<code>LEAN_CROSSWALK.md</code>	Label-complete theorem-to-Lean status map
<code>SHA256SUMS</code>	Integrity hashes for current package contents
<code>data/*.csv</code>	Machine-readable numerical tables
<code>data/numerical_summary.txt</code>	Human-readable verification summary
<code>figures/*.pdf</code>	Vector figures used in the report
<code>figures/*.png</code>	Raster previews of the same figures
<code>audit/CORPUS_AUDIT.md</code>	Claim-level nonduplication audit
<code>audit/SOURCE_AUDIT.md</code>	Source and literature audit
<code>audit/POST_INTAKE_REVIEW.md</code>	Mathematical repairs and replay audit
<code>audit/corpus_manifest_2026-08-27.txt</code>	Preserved recursive TeX path ledger

The trusted mathematical boundary is the prose proof, not the numerical program. Exact rational computations verify finite identities; high-precision values and figures are supporting evidence. External theorems used as inputs are named explicitly in the bibliography and in the status matrix.

References

- [1] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction documentation tree*, GitHub repository, living source inspected 30 August 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>
- [2] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, living LaTeX exposition in the ProveIt repository, 2026. https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
- [3] V. Reshetnikov, *Semi-formalized research frontiers for the Fabius and Rvachev functions*, consolidated living LaTeX volume, 2026. <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers>
- [4] J. Arias de Reyna, *Definición y estudio de una función indefinidamente diferenciable de soporte compacto*, Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales de Madrid **76** (1982), 21–38; English translation, arXiv:1702.05442 (2017).
- [5] J. Arias de Reyna, *Arithmetic of the Fabius function*, Integers **18** (2018), Article A51; arXiv:1702.06487.
- [6] V. A. Rvachev, *Compactly supported solutions of functional-differential equations and their applications*, Russian Mathematical Surveys **45** (1990), no. 1, 87–120. <https://doi.org/10.1070/RM1990v045n01ABEH002324>

- [7] W. Hoeffding, *A class of statistics with asymptotically normal distribution*, Annals of Mathematical Statistics **19** (1948), 293–325. <https://doi.org/10.1214/aoms/1177730196>
- [8] I. M. Sobol', *Sensitivity estimates for nonlinear mathematical models*, Mathematical Modelling and Computational Experiments **1** (1993), 407–414.
- [9] D. Xiu and G. E. Karniadakis, *The Wiener–Askey polynomial chaos for stochastic differential equations*, SIAM Journal on Scientific Computing **24** (2002), 619–644. <https://doi.org/10.1137/S1064827501387826>
- [10] P. Flajolet, X. Gourdon, and P. Dumas, *Mellin transforms and asymptotics: Harmonic sums*, Theoretical Computer Science **144** (1995), 3–58. [https://doi.org/10.1016/0304-3975\(95\)00002-E](https://doi.org/10.1016/0304-3975(95)00002-E)
- [11] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, *On the Lambert W function*, Advances in Computational Mathematics **5** (1996), 329–359. <https://doi.org/10.1007/BF02124750>
- [12] NIST Digital Library of Mathematical Functions, *Chapters 10, 17, and 18: Bessel functions, q -hypergeometric functions, and orthogonal polynomials*, release current at access on 30 August 2026. <https://dlmf.nist.gov/>
- [13] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [14] L. Comtet, *Advanced Combinatorics*, D. Reidel, 1974.
- [15] G. Szegő, *Orthogonal Polynomials*, fourth edition, American Mathematical Society Colloquium Publications, vol. 23, 1975.